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Twenty years since Joinpoint 1.0: Two major enhancements, their justification, and impact

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Since its release of Version 1.0 in 1998, Joinpoint software developed for cancer trend analysis by a team at the US National Cancer Institute has received a considerable attention in the trend analysis community and it became one of most widely used software for trend analysis. The paper published in *Statistics in Medicine* in 2000 (a previous study) describes the permutation test procedure to select the number of joinpoints, and Joinpoint Version 1.0 implemented the permutation procedure as the default model selection method and employed parametric methods for the asymptotic inference of the model parameters. Since then, various updates and extensions have been made in Joinpoint software. In this paper, we review basic features of Joinpoint, summarize important updates of Joinpoint software since its first release in 1998, and provide more information on two major enhancements. More specifically, these enhancements overcome prior limitations in both the accuracy and computational efficiency of previously used methods. The enhancements include: (i) data driven model selection methods which are generally more accurate under a broad range of data settings and more computationally efficient than the permutation test and (ii) the use of the empirical quantile method for construction of confidence intervals for the slope parameters and the location of the joinpoints, which generally provides more accurate coverage than the prior parametric methods used. We show the impact of these changes in cancer trend analysis published by the US National Cancer Institute.

KEYWORDS
change-point, segmented regression, trend analysis

1 | INTRODUCTION

Kim et al[1] proposed to use a segmented line regression model to explain changes in cancer incidence and mortality trends and applied the permutation test to determine the number of segments. The segmented line regression model is a useful simplification of reality, and has gained popularity because the data-driven segments generally have been found to be interpretable, often with respect to the introduction of new cancer control strategies and interventions. The segmented line regression model with continuous trend changes is called a joinpoint regression model and the time-points where the changes occur are called joinpoints in Kim et al.[1] The approach used in Kim et al[1] to determine the number of segments or joinpoints is to sequentially conduct hypothesis tests similarly as in step-wise regression, and the permutation distribution of the test statistic is used due to an intractable analytic distribution of the test statistic. The methods proposed in Kim et al[1] were implemented in Joinpoint software that was developed by a team at the US National Cancer Institute and is available at https://surveillance.cancer.gov/joinpoint/. To fit the segmented line regression model at a given number of segments, the original version of Joinpoint implemented the grid search proposed in Lerman,[2] and Joinpoint allows the user to specify the number of grid points. Inferences on the regression intercept and slope parameters are implemented using the asymptotic results of Feder[3] and Hinkley,[4] and the confidence intervals for the joinpoints are obtained using the likelihood approach. See Kim et al[5] for further details on the inference of the model parameters. Schell[6] used Joinpoint software to identify top articles for applied biostatisticians published between 1985 and 2002 based on citations, and Kim et al[1] was listed as one of the top 66 articles. Joinpoint method and software was originally developed for cancer trend analysis as summarized in the Cancer Statistics Review published by the US National Cancer Institute,[7] and it has been used in various other applications as indicated in the Scopus citation record of 2613 documents (as of July 22 2021) that cited Kim et al.[1] Also, Joinpoint V 4.8 released in April 2020 has been downloaded by 4698 users during the period of April 2020 to March 2021.

Since the release of Joinpoint V 1.0 in 1998, various enhancements and additions have been made to Joinpoint software. Some minor changes include the use of adjusted significance levels in multiple testing in order to improve the detection ability and the use of the t-distribution, instead of the standard normal distribution, in making inferences on the regression slope parameters. A couple of enhanced features include the early stopping that provides a more efficient permutation test without completing the entire set of resampling and the Hudson's continuous fitting algorithm that allows the joinpoints to be estimated anywhere in the time interval, instead of them being limited to specified grid points. Both of these features, early stopping and Hudson's continuous fitting, were implemented in V 3.1 released in 2007, but the Hudson's algorithm was temporarily disabled in V 4.5 due to a technical issue in interface.[8] Also, in V 3.3 released in 2008, the average annual percent change (AAPC) was included as a reported statistic for the average rate of change in a chosen period. As an extension of its capability, a feature to compare two joinpoint models, whether two mean functions are parallel or coincident, was added in V 3.4 (2009) whose details are discussed in Kim et al.[9] Its further extension to cluster the groups with parallel mean functions was discussed in Kim et al,[10] and the addition of clustering algorithm and work on user interface in Joinpoint software is currently on the way. Also, an extended model that allows a jump in the mean function in addition to continuous changes was included in V 4.4, released in 2017. See Chen et al[11] for its motivation, methodology, and examples.

There are two significant and technical changes/additions made in Joinpoint software which overcame prior obstacles by improving the accuracy and efficiency of the model fitting. In order to select the model, that is, the number of joinpoints or segments, the permutation test procedure has been used as the default method since its first release, and then information criteria based approaches have been added as more efficient alternatives to the permutation procedure that is computationally expensive. Computational efficiency has become especially important in the analysis of long term trends of cancer incidence/mortality data collected by the Surveillance, Epidemiology, and the End Results (SEER) program of the US National Cancer Institute, which currently includes annual data from 1975 to 2018, much longer than the period of 1973 to 1995 studied in Kim et al.[1] The Bayesian information criterion (BIC) was added in V 3.0 (2005), for which some details are included in Kim et al,[12] and the modified BIC (MBIC) and BIC_3 that use a harsher penalty than the traditional BIC were added in later versions. These modifications of the BIC performed better when the data had specific characteristics, and Kim et al[13] studies empirical properties of BIC type model selection procedures and proposes two new procedures that internally determine the model selection method based on the data characteristics. These data based model selection methods have been implemented in Joinpoint V 4.6 and V 4.7. Another significant improvement has been made related to the inference of the model parameters. Instead of the asymptotic parametric inference implemented in the original version of Joinpoint following the suggestion in Lerman,[2] which produced quite conservative confidence intervals for the AAPC, the empirical quantile method based on a resampling distribution was proposed to construct a confidence interval for the AAPC (Kim et al;[14] Joinpoint V 4.2 (2015)). The empirical quantile method has been also applied to construct confidence intervals for the annual percent change (APC) and joinpoints in Joinpoint V 4.6 (2018).

Our aims in this article are (i) to briefly summarize the features mentioned above, (ii) to provide more details on the performance of the empirical quantile method applied to make inferences on the APC and the locations of joinpoints, which have not been presented in the papers published to date, and (iii) to present the impact of the updated methods in cancer reporting. In Section 2, we introduce the model and summarize a variety of data types that Joinpoint can handle. Basic concepts and features of Joinpoint in model fitting and inference are reviewed and some extensions are also summarized. Section 3 focuses on two major enhancements: model selection methods based on information criteria and the empirical quantile method in constructing the confidence intervals for the APC and the joinpoints. Applications of joinpoint regression analysis to lung cancer incidence/mortality rates are presented in Section 4. Section 5 includes the impact of the updated methods in cancer reporting. We compare the results of cancer trend data analysis using the current default methods of Joinpoint, the permutation procedure for the model selection and the parametric method for the significance of APC and AAPC values, to those based on updated model selection and inference methods. Concluding remarks are made in Section 6.

2 | MODEL AND FEATURES OF JOINPOINT SOFTWARE

2.1 | Model

Suppose that we observe n pairs of data, $(x_1, y_1), \dots, (x_n, y_n)$, for which the expected value of y given x is represented as a segmented line regression model:

$$\mu(x) = E(y|x) = \begin{cases} \beta_{i,0} + \beta_{i,1}x & \text{for } \tau_{i-1} \leq x < \tau_i \ \ (i = 1, \dots, \kappa), \\ \beta_{\kappa+1,0} + \beta_{\kappa+1,1}x & \text{for } \tau_\kappa \leq x \leq \tau_{\kappa+1}, \end{cases} \tag{1}$$

where $\tau_0 = \min(x_1, \dots, x_n)$ and $\tau_{\kappa+1} = \max(x_1, \dots, x_n)$. If the mean functions are assumed to be continuous at τ_i for $i = 1, \dots, \kappa$, then the continuity constraint is introduced as

$$\beta_{i,0} + \beta_{i,1}\tau_i = \beta_{i+1,0} + \beta_{i+1,1}\tau_i \quad \text{for } \ i = 1, \dots, \kappa.$$

The segmented line regression model with the continuity constraint has been called a broken-line regression model, a joinpoint regression model and so forth. in literature, and we will use "joinpoint regression" in this paper following Kim et al.[1] The time points where the mean function changes, $\tau_1, \dots, \tau_\kappa$, are often called change-points, and it has been called joinpoints in joinpoint regression to emphasize the continuous changes in the mean function. The joinpoint regression model can also be presented in the following form as in Kim et al:[1]

$$\mu(x) = \beta_0 + \beta_1 x + \delta_1(x - \tau_1)^+ + \dots + \delta_\kappa(x - \tau_\kappa)^+, \tag{2}$$

where $a^+ = \max(a, 0)$. Between the models presented in Equations (1) and (2), the parameters are related as follows: $\beta_{1,0} = \beta_0$, $\beta_{1,1} = \beta_1$, and for $i = 2, \dots, \kappa + 1$, $\beta_{i,0} = \beta_0 - \sum_{l=1}^{i-1}\delta_l\tau_l$ and $\beta_{i,1} = \beta_1 + \sum_{l=1}^{i-1}\delta_l$.

In cancer trend analysis, the response (y) is usually the logarithm of the cancer incidence or mortality rate for a given year (x), and we use the continuity constraint unless there is a specific reason to address an abrupt change. The log-transformed rate often better approximates normality, and also facilitates estimation and interpretation of the annual percent change (APC), which for a log-linear model with slope β is defined as $(e^\beta - 1) \times 100$.

For the model presented above, the number of joinpoints, κ, the locations of joinpotins, $(\tau_1, \dots, \tau_\kappa)$, and the regression coefficients, $(\beta_0, \beta_1, \delta_1, \dots, \delta_\kappa)$ are unknown parameters, and our interests are in fitting the model (point estimation and fitting algorithm) and making subsequent inferences (tests and confidence intervals for the model parameters). Sections 2.2 and 2.3 will focus on the inference of the model parameters at given κ and the estimation of κ or equivalently the selection of the model will be discussed in Section 3.1.

2.2 | Model fitting: Point estimation and fitting algorithms

When we assume that κ is known as k and the y_i's are independent with $E(y_i|x_i) = \mu(x_i)$ and $\text{Var}(y_i) = \sigma^2$ $(i = 1, \dots, n)$, the fitting of the model can be done using the ordinary least squares method that finds the values of $\beta_0, \beta_1, \delta_1, \dots, \delta_k, \tau_1, \dots, \tau_k$

by minimizing

$$Q = \sum_{i=1}^{n} (y_i - \mu(x_i))^2$$

and Kim et al[1] applied the grid search method originally proposed by Lerman.[2] The grid search method uses a prespecified grid, say $\mathcal{G}$, for possible locations of the joinpoints and allows the locations of joinpoints, $\{\tau_1, \dots, \tau_k\}$, to be at the grid points, $\{t_1, \dots, t_k\} \subset \mathcal{G}$. Then, the model fitting is done by first finding $(\hat{\beta}_0(\boldsymbol{t}), \hat{\beta}_1(\boldsymbol{t}), \hat{\delta}_1(\boldsymbol{t}), \dots, \hat{\delta}_k(\boldsymbol{t}))$ that minimizes

$$Q(\boldsymbol{t}) = \sum_{i=1}^{n} \left(y_i - \beta_0 - \beta_1 x_i - \delta_1 (x_i - t_1)^+ - \cdots - \delta_k (x_i - t_k)^+ \right)^2$$

at given $\boldsymbol{t} = (t_1, \dots, t_k)'$ and then searching for the value of $\boldsymbol{t}$ that minimizes $Q(\boldsymbol{t})$ over all possible locations of joinpoints, $\boldsymbol{t}$, at the grid. The default of Joinpoint software has been the basic grid search where the given x-values serve as the grid points, and Joinpoint allows a user to specify the "Number of points to place between adjacent observed x values in the grid search." However, the use of a fine grid adds a computational burden, and a method to improve the accuracy of the joinpoint estimation and the computational efficiency of the grid search has been implemented in V 3.1 released in 2007. The Hudson's algorithm[15] added in Joinpoint V 3.1 allows the locations of joinpoints to be anywhere in the data range and it is computationally more efficient than a fine grid search such as the one with three grid points between the consecutive x-values observed. For further discussion on the computational efficiency of the Hudson's algorithm and its superior fitting performance, see Yu et al[16] and Kim et al[5] (Table 1), respectively. However, due to some technical issues in the interface, the Hudson's algorithm was temporarily disabled in V. 4.5, and the efforts to improve the computational efficiency by implementing recently proposed fitting algorithms or improving the Hudson's algorithm are being made. Meanwhile, the fine grid search described above can be used for more accurate fitting than the basic grid search and/or to fill in some specific x-values in a situation like unequally spaced x-values.

As discussed in Kim et al,[1] the weighted least squares (WLS) method can be used when $\text{Var}(y_i) = \sigma_i^2$ $(i = 1, \dots, n)$ and/or the y_i's are not independent. Consider the model presented as:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\boldsymbol{y} = (y_1, \dots, y_n)'$, $\boldsymbol{\beta} = (\beta_0, \beta_1, \delta_1, \dots, \delta_k)'$, $\boldsymbol{\epsilon} = (\epsilon_1, \dots, \epsilon_n)'$,

$$\boldsymbol{X} = \boldsymbol{X}(\boldsymbol{\tau}) = \begin{pmatrix} 1 & x_1 & (x_1 - \tau_1)^+ & \cdots & (x_1 - \tau_k)^+ \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & (x_n - \tau_1)^+ & \cdots & (x_n - \tau_k)^+ \end{pmatrix},$$

and $\text{Cov}(\boldsymbol{y}) = \Sigma$. We assume that $W^{1/2}\Sigma W^{1/2} = \sigma^2 I$ for an appropriately chosen symmetric weight matrix W, where I is the $n \times n$ identity matrix. Then, the weighted least squares estimator of $\boldsymbol{\beta}$ at given $\boldsymbol{t}$ is obtained as

$$\hat{\boldsymbol{\beta}}_w(\boldsymbol{t}) = (\boldsymbol{X}_w'\boldsymbol{X}_w)^{-1}\boldsymbol{X}_w'\boldsymbol{y}_w = (\boldsymbol{X}'W\boldsymbol{X})^{-1}\boldsymbol{X}'W\boldsymbol{y},$$

where $\boldsymbol{y}_w = W^{1/2}\boldsymbol{y}$ and $\boldsymbol{X}_w = \boldsymbol{X}_w(\boldsymbol{t}) = W^{1/2}\boldsymbol{X}(\boldsymbol{t})$, and $\hat{\boldsymbol{\tau}}_w$ can be obtained as the value of $\boldsymbol{t}$ that minimizes $Q_w(\boldsymbol{t}) = \|\boldsymbol{y}_w - \boldsymbol{X}_w(\boldsymbol{t})\hat{\boldsymbol{\beta}}_w(\boldsymbol{t})\|^2$. The final estimator of $\boldsymbol{\beta}$ is $\hat{\boldsymbol{\beta}}_w(\hat{\boldsymbol{\tau}}_w)$. The estimation of $\boldsymbol{\theta} = (\beta_{1,0}, \beta_{1,1}, \dots, \beta_{k+1,0}, \beta_{k+1,1})'$ can be done using the relationship between $\boldsymbol{\theta}$ and $\boldsymbol{\beta}$ described in the previous section, which implies $\boldsymbol{\theta} = A(\boldsymbol{\tau})\boldsymbol{\beta}$ and $\boldsymbol{\beta} = B\boldsymbol{\theta}$ where $A(\boldsymbol{\tau})$ and B are the corresponding $2(k+1) \times (k+2)$ and $(k+2) \times 2(k+1)$ matrices, respectively. For specific presentations of $A(\boldsymbol{\tau})$ and B, see Kim and Kim.[17]

Joinpoint software accommodates various types of the response variable (for example, age-adjusted rate, proportion, Poisson count and so forth) and fits a joinpoint regression model for the observed rates or the logarithm of the rates with the weights either provided or calculated within the program. In most of applications with cancer trend data, the weighted least squares method to handle heteroscedastic errors under the log-linear model has been used, and Joinpoint has a capability of analyzing the data with the autocorrelated errors of lag one, where the autocorrelation parameter is either provided or estimated within the program. When the standard error of the rate, say SE_i for

TABLE 1 Coverage probability of confidence intervals for APC and τ ($\kappa = 1$)

True values				Coverage probabilities								
			10 000	CM		EmpQ1		EmpQ2		CM	EmpQ1	EmpQ2
APC	n	τ	$\times\sigma^2$	APC_1	APC_2	APC_1	APC_2	APC_1	APC_2	τ	τ	τ
(3, 2)	10	3	1	0.781	0.970	0.906	0.886	0.945	0.896	0.980	0.975	0.976
			5	0.876	0.969	0.922	0.893	0.951	0.913	0.978	0.981	0.982
		5	1	0.945	0.964	0.950	0.930	0.958	0.938	0.985	0.980	0.984
			5	0.955	0.968	0.930	0.904	0.943	0.916	0.990	0.993	0.993
		8	1	0.978	0.783	0.899	0.908	0.908	0.946	0.979	0.974	0.978
			5	0.977	0.875	0.901	0.921	0.923	0.948	0.977	0.979	0.982
	20	3	1	0.656	0.960	0.843	0.930	0.933	0.936	0.949	0.965	0.971
			5	0.507	0.960	0.914	0.940	0.964	0.951	0.934	0.972	0.978
		10	1	0.943	0.936	0.960	0.953	0.962	0.957	0.988	0.977	0.979
			5	0.904	0.920	0.968	0.957	0.968	0.959	0.958	0.982	0.984
		17	1	0.960	0.843	0.936	0.947	0.941	0.969	0.979	0.985	0.984
			5	0.960	0.643	0.938	0.960	0.946	0.979	0.959	0.990	0.992
	30	3	1	0.634	0.953	0.789	0.946	0.896	0.950	0.950	0.953	0.961
			5	0.437	0.959	0.892	0.957	0.953	0.963	0.920	0.962	0.974
		15	1	0.936	0.947	0.949	0.955	0.957	0.961	0.997	0.980	0.983
			5	0.919	0.924	0.953	0.955	0.957	0.956	0.968	0.968	0.971
		27	1	0.945	0.833	0.941	0.935	0.944	0.954	0.975	0.975	0.975
			5	0.952	0.576	0.946	0.940	0.950	0.971	0.940	0.979	0.982
(3, 0)	10	3	1	0.937	0.966	0.901	0.922	0.935	0.928	0.996	0.980	0.980
			5	0.819	0.970	0.896	0.890	0.936	0.896	0.983	0.975	0.975
		5	1	0.960	0.963	0.936	0.929	0.953	0.949	0.999	0.982	0.989
			5	0.942	0.962	0.941	0.937	0.951	0.942	0.986	0.976	0.978
		8	1	0.973	0.937	0.931	0.902	0.939	0.937	0.997	0.980	0.981
			5	0.977	0.809	0.899	0.897	0.909	0.935	0.983	0.971	0.972
	20	3	1	0.938	0.963	0.933	0.952	0.962	0.957	0.998	0.988	0.984
			5	0.761	0.958	0.854	0.929	0.921	0.933	0.967	0.965	0.965
		10	1	0.955	0.949	0.957	0.949	0.967	0.960	1.000	0.991	0.996
			5	0.949	0.945	0.956	0.950	0.962	0.959	0.995	0.978	0.982
		17	1	0.954	0.949	0.953	0.964	0.962	0.970	1.000	0.991	0.992
			5	0.954	0.896	0.946	0.955	0.953	0.965	0.988	0.984	0.984
	30	3	1	0.942	0.957	0.938	0.955	0.959	0.960	0.997	0.990	0.987
			5	0.763	0.954	0.843	0.945	0.892	0.948	0.969	0.958	0.963
		15	1	0.965	0.967	0.959	0.962	0.966	0.969	1.000	0.989	0.994
			5	0.942	0.947	0.953	0.956	0.962	0.964	0.999	0.986	0.989
		27	1	0.951	0.938	0.952	0.959	0.958	0.963	0.999	0.993	0.993
			5	0.946	0.892	0.949	0.956	0.955	0.963	0.986	0.980	0.978

Note: CM is the parametric method that is the current default of Joinpoint software, and EmpQ1 and EmpQ2 denote the empirical quantile methods based on u' and (4), respectively. The mean function is $\mu(x) = \beta_{1,0} + \beta_{1,1}x + (\beta_{2,1} - \beta_{1,1})(x - \tau)^+$, where $a^+ = \max(a, 0)$, $\beta_{1,0} = 1$, $\beta_{1,1} = \log(1 + \text{APC}_1/100)$, $\beta_{2,1} = \log(1 + \text{APC}_2/100)$, and the errors are independent and normally distributed with mean zero and variance σ^2.

the ith response, is provided such as the one from SEER*Stat software developed by the US National Cancer Institute[18] and the correlations among the observed rates are negligible, the weight matrix $W = (w_{ij})$ is defined as the one with $w_{ij} = 0$ for $i \neq j$ and $w_{ii} = 1/(\mathrm{SE}_i)^2$ $(i, j = 1, \dots, n)$ for the linear model such that $E(y_i|x_i) = \mu(x_i)$. If one uses the log-linear model for the rate r such that $E(\log(r_i)|x_i) = \mu(x_i)$, then Joinpoint estimates the weight as $w_{ii} = r_i^2/(\mathrm{SE}_i)^2$ using the first-order approximation for the variance of $\log(r_i)$. When a user asks Joinpoint to calculate the response variable such as crude rates, age-adjusted rates, proportions and percentages, an appropriate form of standard errors will be calculated within the program. Joinpoint V 4.9.0.0 released in March 2021 accommodates data with a general covariance structure, and a user is supposed to provide a covariance matrix. More details on the types of data that can be provided by a user or requested to be calculated, along with how the standard errors are estimated, can be found at the Joinpoint help site.[19] Joinpoint reports the estimated model parameters including the intercept and slope for each segment, the sizes of slope changes, the locations of joinpoints, and the APC values. With the WLS fit, the model sum of squared errors is calculated as $Q_w(\hat{\boldsymbol{\tau}}) = \boldsymbol{y}_w'\{I - \boldsymbol{X}_w(\hat{\boldsymbol{\tau}})(\boldsymbol{X}_w'(\hat{\boldsymbol{\tau}})\boldsymbol{X}_w(\hat{\boldsymbol{\tau}}))^{-1}\boldsymbol{X}_w'(\hat{\boldsymbol{\tau}})\}\boldsymbol{y}_w$, which is also reported in the Joinpoint output.

One summary statistic added in Joinpoint V 3.3 is the average annual percent change (AAPC) that summarizes the average rate of changes over a fixed time period of our interest, which can be composed with more than one segment in the fit made by joinpoint regression. As described in the Joinpoint help site[20] and in Clegg et al,[21] "it allows us to use a single number to describe the average APCs over a period of multiple years." The AAPC is based on a weighted average of the slope parameters and the estimated AAPC is formally defined as

$$\widehat{AAPC} = \left\{ \exp\left(\frac{\sum_i d_i \hat{\beta}_{i,1}}{\sum_i d_i} \right) - 1 \right\} \times 100,$$

where the sum is over the segments in the preselected period, d_i denotes the length of each segment in the time range of interest, and $\hat{\beta}_{i,1}$ is the estimated slope parameter for the ith segment.

2.3 | Inference: Tests and confidence intervals

Inferences about the regression coefficients have been performed using the asymptotic and parametric results such as those of Feder,[3] Hinkley,[4,22] and Kim and Kim.[23] For the joinpoint model described in (1) and (2), let $\boldsymbol{\beta} = (\beta_0, \beta_1, \delta_1, \dots, \delta_k)'$ and $\boldsymbol{\theta} = (\beta_{1,0}, \beta_{1,1}, \dots, \beta_{k+1,0}, \beta_{k+1,1})'$. Then, under some conditions on the distribution of ϵ and the spacing of $\boldsymbol{x}$, the least squares estimator of $\boldsymbol{\theta}$, $\hat{\boldsymbol{\theta}}$, is approximately normally distributed with mean $\boldsymbol{\theta}$ and variance $\sigma^2\Sigma_u$, where $\Sigma_u = (\boldsymbol{X}_u'(\boldsymbol{\tau})\boldsymbol{X}_u(\boldsymbol{\tau}))^{-1}$ with the block diagonal matrix $\boldsymbol{X}_u$ whose block is the regression matrix for the corresponding segment without the continuity constraint. For $k = 2$, for example,

$$X_u(\boldsymbol{\tau}) = \begin{pmatrix} 1 & x_1 & 0 & 0 & 0 & 0 \\ 1 & x_2 & 0 & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & x_{j_1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & x_{j_1+1} & 0 & 0 \\ 0 & 0 & 1 & x_{j_1+2} & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 1 & x_{j_2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & x_{j_2+1} \\ 0 & 0 & 0 & 0 & 1 & x_{j_2+2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 1 & x_n \end{pmatrix},$$

where $x_{j_1} \le \tau_1 < x_{j_1+1}$ and $x_{j_2} \le \tau_2 < x_{j_2+1}$. Then, we can estimate the variance of $\hat{\boldsymbol{\theta}}$ as $\hat{\sigma}^2\hat{\Sigma}_u$, where $\hat{\Sigma}_u = (\boldsymbol{X}_u'(\hat{\boldsymbol{\tau}})\boldsymbol{X}_u(\hat{\boldsymbol{\tau}}))^{-1}$ and $\hat{\sigma}^2 = \|\boldsymbol{y} - \boldsymbol{X}_u(\hat{\boldsymbol{\tau}})\hat{\boldsymbol{\theta}}\|/(n-3k-2)$, and the variance of $\hat{\boldsymbol{\beta}}$ can be obtained as $\hat{\Lambda}_u = \hat{\sigma}^2 B\hat{\Sigma}_u B'$ where B is the $(k+2)\times 2(k+1)$ matrix such that $\boldsymbol{\beta} = B\boldsymbol{\theta}$. Note that the asymptotic variance of $\hat{\boldsymbol{\theta}}$ is same as the asymptotic variance of the estimated regression coefficients in a multiphase regression model without the continuity constraint, and more discussion on this issue can be found in Kim and Kim.[17] In calculating the asymptotic standard error of the estimated regression coefficients, however, there is an ambiguity on which block diagonal matrix an observed x should be included when the x value coincides with the estimated joinpoint. Lerman[2] suggested to delete such offending observations, the data points that coincide with the estimated joinpoints, in order to avoid such ambiguity, and Joinpoint followed this suggestion by deleting the offending observations to estimate $\hat{\Sigma}_u$. Some simulation results on the effects of keeping the offending observations can be found in Kim et al,[5] which empirically supports the practice to remove the offending observations. To estimate σ^2, Joinpoint has used the weighted residual sum of squares under the unconstrained model fit divided by the degrees of freedom to estimate $\mathrm{Cov}(\hat{\boldsymbol{\theta}})$, $n-3k-2$ for the unconstrained model.

In the early version of Joinpoint, the normal distribution was used to find a confidence interval of a regression coefficient or the P-value of the test for a regression coefficient being different from zero, and the t-distribution has been used from V 2.6 to better accommodate small sample behavior of the sample statistics. See Kim et al[5] for further details regarding small sample performances of these asymptotic confidence intervals as well as their robustness. For the inference on the AAPC, however, Joinpoint uses the normal distribution unless the time period of interest is composed with one segment, considering that the estimated AAPC is a function of a weighted average of the estimated slope coefficients that are asymptotically normally distributed. As discussed in Clegg et al[21] and Kim et al,[14] the AAPC confidence interval based on the normal distribution tends to be conservative with the coverage probability usually larger than the nominal level, and Kim et al[14] proposed the empirical quantile method to construct a confidence interval for the AAPC. The empirical quantile method is a resampling based method, and its superior performance has been demonstrated in Kim et al.[14] The empirical quantile method for the AAPC was implemented in V 4.2 (2015), and some details can also be found at the Joinpoint help site.[24]

For the locations of joinpoints, $\boldsymbol{\tau}$, the convergence of $\hat{\boldsymbol{\tau}}$ to the normal distribution is known to be rather slow[22] although $\hat{\boldsymbol{\tau}}$ is also $\sqrt{n}$-consistent as shown in Feder[3] and Kim and Kim,[23] and we have used the likelihood method to obtain a confidence interval for the joinpoint as discussed in Kim et al,[5] which is also briefly explained at the Joinpoint help site.[25] Motivated by the nonparametric nature of the empirical quantile method and satisfactory performance in constructing the confidence interval of AAPC, we applied the empirical quantile method to calculate the confidence limits for APC values and the locations of joinpoints, and we will summarize its details in Section 3.

2.4 | Extensions

Joinpoint released V 4.9.0.0 in March, 2021, and the current version includes various enhancements and updates since its first version, V 1.0. (1998) as discussed in the previous sections. In addition to technical and interface changes described above, some major technical extensions made and to be made include the comparability test added in V 3.4 (2009) whose details are discussed in Kim et al,[9] clustering of joinpoint models proposed in Kim et al,[10] and jump model analysis discussed in Chen et al.[11] The comparability test was developed to determine if two groups share the same mean functions (test of coincidence) or their mean functions are parallel (test of parallelism), where the mean function is represented as a joinpoint regression mean function. Tests of coincidence and parallelism use the conventional F-type statistic and estimate its P-value using the permutation distribution obtained by permuting the pairwise residuals at each time-point obtained under the null hypothesis. Extending the idea of the two-group comparability test, we developed a procedure to cluster groups of cohorts into several clusters where the groups in each cluster share the common or parallel mean functions. The method of clustering the ordered groups and simulation results are presented in Kim et al,[10] and it will be available in a future version of Joinpoint. Kim et al[10] considers the situation where the groups are ordered such as the consecutive 19 age groups used in cancer incidence and mortality trend analysis, and the proposed method performs the clustering of the groups for a given number of clusters via exhaustive searches and estimates the number of clusters using the permutation test and BIC. The jump model was proposed as a way to address a coding change in international classification of diseases (ICD) for cancer staging, and the model allows a jump in the mean function at a known time point in addition to continuous changes explained by the joinpoint regression model. Its algorithm was implemented in V 4.4 (2017) and Chen et al[11] includes a discussion on the method and examples. A brief summary on each extension can be found at the Joinpoint help site.[26]

3 | TWO MAJOR ENHANCEMENTS IN MODEL SELECTION AND INFERENCE FOR APC AND JOINPOINTS

3.1 | Model selection

For the discussion of the model fitting and inference in Sections 2.2 and 2.3, we considered the model with the number of joinpoints fixed as k, but the number of joinpoints, κ, is usually unknown in practice and its estimation is an important problem in trend analysis. Kim et al[1] proposed to use the permutation test procedure to determine the number of joinpoints, κ, by sequentially applying the test of the null hypothesis that there are k_0 joinpoints against the alternative hypothesis of k_1 joinpoints ($k_0 < k_1$). The procedure starts with $k_0 = \text{JP}_{\min}$ and $k_1 = \text{JP}_{\max}$ with predetermined values of $\text{JP}_{\min}$ and $\text{JP}_{\max}$ where $\text{JP}_{\min} < \text{JP}_{\max}$, and proceeds to the test of $k_0 + 1$ vs k_1 if the null hypothesis is rejected, and to the test of k_0 vs $k_1 - 1$ joinpoints otherwise. Because the null distribution of the F-type statistic does not follow a classical asymptotic theory, the permutation distribution of the test statistic has been used to estimate the P-value of the test. We repeat such permutation tests until $k_0 = k_1$. The significance level of each permutation test of $H_0 : k_0$ joinpoints vs $H_a : k_1$ joinpoints, starting with the test of $\text{JP}_{\min}$ joinpoints vs $\text{JP}_{\max}$ joinpoints, is adjusted to address the multiple testing being conducted. In earlier versions, $\alpha/(\text{JP}_{\max} - \text{JP}_{\min})$ has been used, where α is the overall significance level, and in a later version, the significance level of the permutation test for the null hypothesis of k_0 joinpoints was set at $\alpha/(\text{JP}_{\max} - k_0)$, which controls the over-fitting probability under α (ie, $P(\hat{\kappa} > \kappa^*) \leq \alpha$, where $\hat{\kappa}$ is an estimator of κ and κ^* is the true number of joinpoints) and improves the power of the test compared to the previously used values of the significance level. See Kim et al[12] for further details. Also, as a way to improve the computational efficiency of the permutation test, a sequential stopping was added in V 3.1. The early stopping option allows the resampling to stop before it completes the entire N enumerations originally set by a user (or the default value of 4499) if there is sufficient evidence for the final P-value being either smaller or larger than the significance level. For further details on the early stopping, please see Fay et al[27] and Kim.[28] A brief summary on the permutation test and the early stopping can also be found at the Joinpoint help site.[29]

Another frequently used model selection method is the one based on information-based criteria such as Akaike information criterion (AIC) of Akaike[30] and Bayes (or Bayesian) information criterion proposed by Schwarz.[31] These information based criteria typically include two parts: one part that indicates the accuracy of the fit and the other part that penalizes for the complexity of the model. The traditional forms of AIC and BIC are as follows:

$$\text{AIC}(k) = -2\log L(\hat{\boldsymbol{\theta}}) + 2p_k,$$
$$\text{BIC}(k) = -2\log L(\hat{\boldsymbol{\theta}}) + p_k \log(n),$$

where $L(\hat{\boldsymbol{\theta}})$ is the maximized likelihood function and p_k denotes the dimensionality of the kth model. For a segmented line regression model with k-joinpoints and the normally distributed errors with unknown variance, these are equivalent to

$$\text{AIC}(k) = \log\left(\frac{\text{RSS}(k)}{n}\right) + \frac{2p_k}{n},$$
$$\text{BIC}(k) = \log\left(\frac{\text{RSS}(k)}{n}\right) + p_k\frac{\log(n)}{n},$$

where $\text{RSS}(k)$ is the residual sum of squares for the fit of the model with k-joinpoints, $p_k = 2k + 2$ or equivalently $p_k = 2k$, and the model that minimizes these selection measures is selected as a final model. In typical regression model selection settings, BIC is known to be a consistent model selection tool if the set of candidate models includes the true model, and also our empirical study has indicated a more liberal tendency of AIC, that is, a higher probability of over-fitting the model, than BIC. Considering that our goal in cancer trend analysis is often to choose a parsimonious model, we implemented BIC in V 3.0 (2005) as a method to estimate κ. BIC in the context of joinpoint regression was discussed in Kim et al,[12] where the traditional BIC was shown to be more liberal than the permutation test with the overall significance level of 0.05. Other BIC type measures have been introduced in later versions of Joinpoint: a modified BIC in V 3.5 (2011) following Zhang and Siegmund[32] who proposed a modification of BIC in the context of change-point problems and a type of BIC measure with a harsher penalty than the traditional BIC, BIC_3 with $p_k = 3k$, in V 4.6 (2018). Also in V 4.6 (2018), a data driven selection method (DDS) was added as an alpha feature where one of BIC or BIC_3 is internally selected as a model selection method based on the data characteristics, and V 4.7 (2019) includes an additional data-driven selection method, the weighted BIC (WBIC), that combines BIC and BIC_3 with the weight estimated from the data. These procedures are motivated from a need to guide a user to choose a selection method that performs better for a given data set and are developed based on an empirical study where BIC performs better to identify changes with small effect sizes, slope change sizes adjusted to the variability in data, while the modified BIC and BIC_3 perform better when the effect size is relatively large. More discussion and details on these selection procedures can be found in Kim et al[13] and a brief description is provided at the Joinpoint help site.[33]

Because the inferences reviewed in Section 2 are conditional on the number of joinpoints κ, it is important to accurately estimate κ. For the permutation test procedure that was the selection method originally proposed in Kim et al,[1] Kim et al[12] showed that the consistency of $\hat{\kappa}$ can be achieved by appropriately choosing α. Kim and Kim[34] studied the consistency of BIC type selection measures in segmented line regression, and showed that BIC_4 with $p_k = 4k$ is a consistent model selection procedure for a model without the continuity constraint and provided empirical evidence for the consistency of BIC_3 for the joinpoint regression model. Although not reported in this paper, our simulation studies indicated satisfactory large sample behavior of WBIC in correctly selecting κ.

Remark. The choice of JP_{min} and JP_{max} affects the performance of the model selection. For the permutation test procedure, the significance level at each step of the multiple tests decreases as JP_{max} increases, which is expected to lead to a loss of power in general. However, its effect is more complex with multiple testing, for example, different sets of null and alternative hypotheses tested at the steps of multiple testing. With WBIC, which is being considered as a new default model selection method of Joinpoint software, its effect was observed to be small in our empirical study and more details will be provided in a separate paper. A practical guideline based on the length of the data is given in the Joinpoint help site,[35] and Joinpoint follows this guideline for the default setting of JP_{min} and JP_{max}.

3.2 | Empirical quantile based inference for AAPC, APC, and Joinpoints

3.2.1 | Methods

As discussed in Sections 2.2 and 2.3, AAPC was included as a reported statistic from V 3.3, and the earlier version of Joinpoint used the asymptotic normal distribution to construct a confidence interval for the true AAPC (Clegg et al[21]). Noting a conservative tendency of the confidence interval suggested in Clegg et al,[21] Muggeo[36] proposed an improved confidence interval for the AAPC that incorporates the joint distribution of the estimated slope parameters and joinpoint locations. Kim et al[14] proposed a new method to construct a confidence interval for the true AAPC using a resampling distribution, which was called the empirical quantile method and has been implemented in Joinpoint V 4.2. Extensive simulations have been conducted to assess the accuracy of the empirical quantile method, and the results presented in Kim et al[14] indicate that the proposed method performs better than the parametric method of Muggeo.[36]

Joinpoint V 4.6 applied the empirical quantile method to construct confidence intervals for the APC and joinpoint parameters as well, and we summarize their performances in this section. For the joinpoint regression model (1) or (2) where $\mu(x) = E(\log r|x)$, the annual percent change (APC) for the jth segment with the slope of β_j is defined as $APC_j = \left\{\exp(\beta_j) - 1\right\} \times 100$, where $\beta_j = \beta_{j,1} = \beta_1 + \sum_{l=1}^{j-1}\delta_l$ for $j = 2, \dots, \kappa + 1$. Suppose that Joinpoint software selects the model with k joinpoints as the final model (ie, $\hat{\kappa} = k$), and one wishes to find a confidence interval for the APC of the jth segment. The parametric method in Joinpoint software calculates the $100(1-\alpha)\%$ confidence limits for APC_j as follows:

$$
\begin{aligned}
APC_{j,L(\alpha)} &= \left\{\exp\left[\hat{\beta}_j - t_{1-\alpha/2,df}SE(\hat{\beta}_j)\right] - 1\right\} \times 100, \\
APC_{j,U(\alpha)} &= \left\{\exp\left[\hat{\beta}_j + t_{1-\alpha/2,df}SE(\hat{\beta}_j)\right] - 1\right\} \times 100,
\end{aligned} \tag{3}
$$

where $t_{p,d}$ is the pth percentile of the t-distribution with d degrees of freedom and $SE(\hat{\beta}_j)$ is estimated by using the information matrix based on the unconstrained model with offending observations deleted. In this section, we describe how to construct a confidence interval using the empirical cumulative distribution function (CDF) of the residuals and call it the empirical quantile confidence interval. Note that this is the method proposed and applied to the AAPC in Kim et al[14] and we use the same notations and descriptions of Kim et al[14] except those in Step 5 and Step 6 below.

- Step 1: For the least squares fit of the k joinpoint model, calculate the residuals, $\hat{\epsilon}_i = y_i - \hat{y}_i$, where $\hat{y}_i$ is obtained with the model with k joinpoints.
- Step 2: Generate B independent samples of size n from a uniform distribution on $(0, 1)$. That is, for each $b = 1, \dots, B$, generate $\boldsymbol{U}^{(b)} = (u_1, \dots, u_n)'$, where the u_i's are iid from $U(0, 1)$.
- Step 3: For $b = 1, \dots, B$, generate $\boldsymbol{\epsilon}^{(b)} = F_n^{-1}(\boldsymbol{U}^{(b)})$, where F_n denotes the empirical distribution function of the residuals, $\hat{\boldsymbol{\epsilon}} = (\hat{\epsilon}_1, \dots, \hat{\epsilon}_n)'$ with $\hat{\epsilon}_i = y_i - \hat{y}_i$, and F_n and F_n^{-1} are described below.
- Step 4: For $b = 1, \dots, B$, generate the bth resample data, $\boldsymbol{y}^{(b)} = \hat{\boldsymbol{y}} + \boldsymbol{\epsilon}^{(b)}$, where $\hat{\boldsymbol{y}}$ is the least squares fit of the original data for the k joinpoint model.
- Step 5: For each $b = 1, \dots, B$, fit the model (1) with k joinpoints for $\boldsymbol{y}^{(b)}$, the bth resample data, and estimate the slope parameters for the $k + 1$ linear segments. Denote the slopes estimated with the bth resample data for the $(k + 1)$ segments in the k-joinpoint model as $(\hat{\beta}_1^{(b)}, \dots, \hat{\beta}_{k+1}^{(b)})$. Note that we let $\beta_{j,1} = \beta_j$ $(j = 1, \dots, k + 1)$ in this section for notational simplicity and thus $(\hat{\beta}_1^{(b)}, \dots, \hat{\beta}_{k+1}^{(b)}) = (\hat{\beta}_{1,1}^{(b)}, \dots, \hat{\beta}_{k+1,1}^{(b)})$.
- Step 6: Construct the empirical CDF confidence limits for APC_j as follows:

$$\text{APC}_{j,L(\alpha)} = \left\{ \exp(\hat{\beta}^B_{j,\alpha/2}) - 1 \right\} \times 100,$$
$$\text{APC}_{j,U(\alpha)} = \left\{ \exp(\hat{\beta}^B_{j,1-\alpha/2}) - 1 \right\} \times 100,$$

where $\hat{\beta}^B_{j,p}$ is the 100pth empirical percentile of $\{\hat{\beta}_j^{(1)}, \dots, \hat{\beta}_j^{(B)}\}$, the set of the jth segment slope estimates obtained from B-many resamples.

In Step 3, we use the following definition of truncated F_n and F_n^{-1} similarly as in Kim et al.[14] For the ordered residuals, $\hat{\epsilon}_{(1)} \le \cdots \le \hat{\epsilon}_{(n)}$, define the end points of the intervals where F_n is constant as $z_{(0)}, z_{(1)}, \dots, z_{(n+1)}$ where $z_{(i)} = \hat{\epsilon}_{(i)}$ for $i = 1, \dots, n$, and $z_{(0)}$ and $z_{(n+1)}$ are defined as follows: For $D = 1.5 \times IQR$, where $IQR =$ third quartile of $\hat{\boldsymbol{\epsilon}}$ − first quartile of $\hat{\boldsymbol{\epsilon}} = Q_3 - Q_1$.

$$z_{(0)} = \begin{cases} \hat{\epsilon}_{(1)} - \frac{2}{n} \times IQR & \text{if } \hat{\epsilon}_{(1)} \le Q_1 - D, \\ Q_1 - D & \text{otherwise,} \end{cases}$$
$$z_{(n+1)} = \begin{cases} \hat{\epsilon}_{(n)} + \frac{2}{n} \times IQR & \text{if } \hat{\epsilon}_{(n)} \ge Q_3 + D, \\ Q_3 + D & \text{otherwise.} \end{cases}$$

Then, define F_n and F_n^{-1} as follows:

$$F_n(t) = (n+1)^{-1} \sum_{i=1}^{n+1} I(z_{(i)} \le t),$$
$$F_n^{-1}(u) = \begin{cases} z_{(i)} + \left(z_{(i+1)} - z_{(i)}\right) * \left(u - \frac{i}{n+1}\right) * (n+1), & \text{if } u \in \left[\frac{i}{n+1}, \frac{i+1}{n+1}\right) \text{ for } i \in \{0, \dots, n\}, \\ z_{(n+1)}, & \text{if } u = 1. \end{cases} \tag{4}$$

Remark 1. The choice of $z_{(0)}$ and $z_{(n+1)}$, which is a modification of the one in Kim et al,[14] is proposed to obtain more robust values of the resampled residuals.

Remark 2. Kim et al[14] discussed two different definitions of $F_n^{-1}(u)$: one is (4) described above and the other uses $F_n^{-1}(u) = z_{(i)} + \left(z_{(i+1)} - z_{(i)}\right) * u'$ for $u \in \left[\frac{i}{n+1}, \frac{i+1}{n+1}\right)$, where u and u' are independent random numbers from the uniform distribution on $(0, 1)$. Since our simulations indicated that the use of $F_n^{-1}(u)$ defined in (4) showed a better performance of the confidence interval for APC, we used (4) in the main description above, but empirical studies will be presented for both methods: we call the method based on $F_n^{-1}(u) = z_{(i)} + \left(z_{(i+1)} - z_{(i)}\right) * u'$ as "EmpQ1" and the method based on (4) as "EmpQ2".

Remark 3. When the confidence interval for APC is calculated by the empirical quantile method, the resulting confidence interval is conditional only on the number of segments and should be interpreted as a confidence interval for the APC of a segment such as the last segment, second segment and so forth, not conditional on the estimated joinpoints.

Regarding the locations of joinpoints, a parametric $100(1-\alpha)\%$ confidence set for $\boldsymbol{\tau} = (\tau_1, \dots, \tau_k)$ for a model with k joinpoints is constructed as

$$\left\{ \boldsymbol{t} = (t_1, \dots, t_k) : Q_k(\boldsymbol{t}) \leq Q_k(\hat{\boldsymbol{\tau}}) \left(1 + \frac{k}{n-d} F^{-1}_{k,n-d}(1-\alpha) \right) \right\}, \tag{5}$$

where $d = n - (2k+2)$, $F^{-1}_{\nu_1,\nu_2}(p)$ is the pth percentile of an F-distribution with ν_1 and ν_2 degrees of freedom, $Q_k(\boldsymbol{t})$ is the residual sum of squares for the segmented line regression model with changes at $\boldsymbol{t}$, and $Q_k(\hat{\boldsymbol{\tau}})$ is the residual sum of squares with $\hat{\boldsymbol{\tau}}$ (See Kim et al[5]). Now, we describe how to construct a confidence interval for τ_j ($j = 1, \dots, k$) using the empirical CDF of the residuals.

- Step 1 to Step 4: Follow the same steps described above for the APC confidence interval.
- Step 5: For each $b = 1, \dots, B$, fit the model (1) with k joinpoints for $\boldsymbol{y}^{(b)}$, the bth resample data, and estimate $\hat{\tau}_1^{(b)}, \dots, \hat{\tau}_k^{(b)}$.
- Step 6: For $j = 1, \dots, k$, construct the empirical CDF confidence limits for τ_j as $\tau_{j,L(\alpha)} = \hat{\tau}^B_{j,\alpha/2}$ and $\tau_{j,U(\alpha)} = \hat{\tau}^B_{j,1-\alpha/2}$, where $\hat{\tau}^B_{j,p}$ is the 100pth empirical percentile of $\{\hat{\tau}_j^{(1)}, \dots, \hat{\tau}_j^{(B)}\}$.

3.2.2 | Simulations

We conducted a simulation study to assess the performance of the empirical quantile confidence intervals for APC and τ. As a performance measure, the coverage probabilities of the 95% confidence intervals for the APC parameters are estimated for three methods: the current parametric method based on the t-interval in (3)(CM), the empirical quantile method with u' (EmpQ1/EQ1), and the empirical quantile method with u in (4) (EmpQ2/EQ2). Differently from the inference on AAPC, the confidence interval for APC is based on one segment, and thus the correlations between the estimated regression slopes and joinpoints do not play a role as in the parametric method of Muggeo.[36] Although inference on APC is made for each segment, the empirical quantile confidence interval for the APC parameter incorporates the variability of estimated joinpoints, while the current parametric method uses the standard error estimated conditional on the estimated joinpoint locations. For the estimation of the joinpoint locations $\boldsymbol{\tau}$, CM denotes the parametric method based on the likelihood function (5), and EmpQ1 and EmpQ2 are the empirical quantile methods described above. We used the responses generated under the normal distribution with mean $\mu(x)$ in (2) for various settings of κ, n, $\boldsymbol{\tau}$, $\boldsymbol{\beta}$, and σ^2 where β_0 is set as 1, and the number of simulations was 5000. The results for 108 cases with $\kappa = 1$ and 120 cases with $\kappa = 2$ are summarized by n in Figures 1 and 2, and some of the results are listed in Tables 1 and 2. One hundred eight different settings of $n, \tau, \beta_1, \delta_1$, and σ^2 are used when $\kappa = 1$, and a part of these 108 cases, 36 cases, are presented in Table 1. Similarly, 120 different settings of $n, \tau_1, \tau_2, \beta_1, \delta_1, \delta_2$, and σ^2 are considered when $\kappa = 2$ and 24 cases are presented in Table 2.

Box-plots in Figures 1 and 2 are obtained for the 108 cases simulated under $\kappa = 1$ ($n = 10, 20, 30$) and 120 cases under $\kappa = 2$ ($n = 10, 30, 50$). In these figures, the coverage probabilities of the 95% confidence intervals for the true APC values, the proportion of simulated data sets whose 95% confidence intervals contain the true parameter value, are shown via box-plots. In this article, we report the results based on the yearly grid search only although we also conducted simulations with the quarterly grid search. A different choice of the grid indicated some minor changes in the coverage probabilities, but their overall differences do not seem to be big enough to require the presentation of the results for the both grids. Thus, all the results presented in Tables 1 and 2 and Figures 1 and 2 are the ones based on the yearly grid search. In order to make comparisons easier, the plots in Figures 1 and 2 only show the cases whose coverage probabilities are greater than the minimum value in the vertical axis, 0.7 in Figures 1 and 2. The numbers of extreme cases whose coverage probabilities are lower than 0.7 and thus not shown in the plots are marked in the parentheses, just above the horizontal axis.

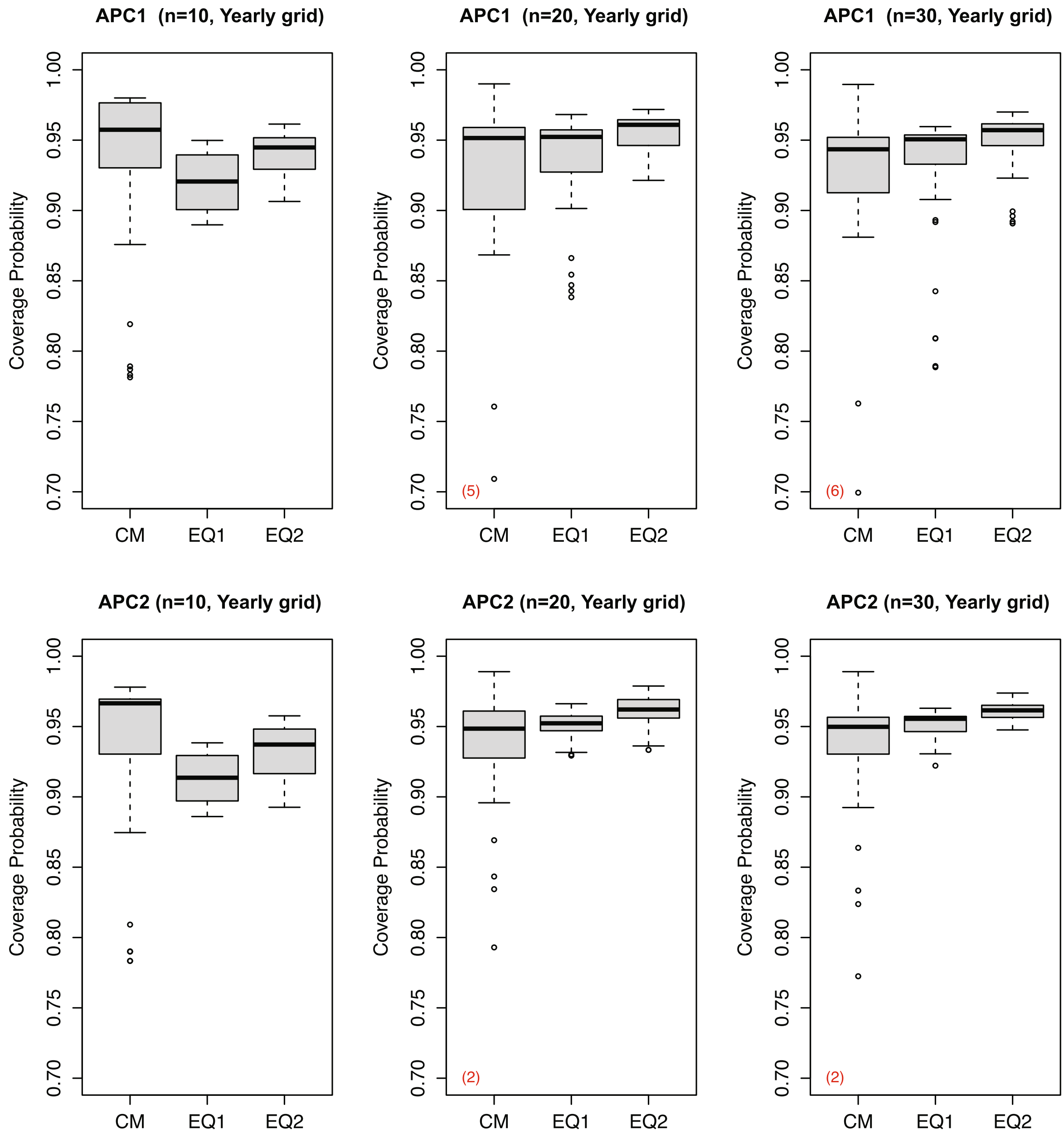


FIGURE 1 Box plots of coverage probability of APC confidence intervals by n: $\kappa = 1$, 36 cases for each of $n = 10, 20, 30$ (CM is the parametric method that is the current default of Joinpoint software, and EQ1 and EQ2 denote the empirical quantile methods based on u' and (4), respectively)

A most prominent difference between the parametric confidence interval and the empirical quantile based intervals observed in Figures 1 and 2 is that the coverage probabilities of the confidence interval constructed by parametric method (CM) vary a lot more than those by the empirical quantile method (EQ1 and EQ2). We observe in Figures 1 and 2 that the EmpQ1 confidence interval has a lower coverage probability than the EmpQ2 confidence interval in general and the EmpQ1 method has a tendency of underestimating the coverage probability when n is as small as 10 while the EmpQ2 method tends to be more conservative as n increases. It is also observed that the performance of the EmpQ2 method is more robust in terms of the coverage probability. We also conducted a simulation study to compare the performance of the confidence intervals for the joinpoint locations, where the parametric confidence interval and the empirical quantile confidence interval were observed to perform comparably. Both methods tend to produce conservative confidence intervals for the locations of joinpoints, and its conservative tendency still remained with the quarterly grid search.

TABLE 2 Coverage probability of confidence intervals for APC and τ ($\kappa = 2$)

True values				**Coverage probabilities**														
			10 000	**CM**			**EmpQ1**			**EmpQ2**			**CM**		**EmpQ1**		**EmpQ2**	
APC	n	(τ_1, τ_2)	$\times\sigma^2$	APC_1	APC_2	APC_3	APC_1	APC_2	APC_3	APC_1	APC_2	APC_3	τ_1	τ_2	τ_1	τ_2	τ_1	τ_2
(1, 3, 2)	10	(3, 8)	1	0.958	0.990	0.971	0.888	0.826	0.915	0.918	0.857	0.939	0.999	1.000	0.982	0.993	0.990	0.994
			5	0.971	0.989	0.976	0.898	0.847	0.884	0.924	0.886	0.911	1.000	1.000	0.994	0.996	0.996	0.997
		(4, 7)	1	0.974	0.987	0.979	0.913	0.930	0.880	0.943	0.950	0.893	1.000	1.000	0.985	0.982	0.989	0.988
			5	0.976	0.987	0.976	0.887	0.899	0.865	0.924	0.923	0.890	1.000	1.000	0.994	0.992	0.995	0.994
	20	(3, 18)	1	0.823	0.970	0.665	0.860	0.799	0.873	0.886	0.828	0.929	0.993	0.984	0.955	0.935	0.978	0.943
			5	0.639	0.978	0.643	0.902	0.870	0.957	0.936	0.904	0.967	0.985	0.981	0.970	0.961	0.983	0.969
		(7, 14)	1	0.943	0.947	0.934	0.940	0.942	0.955	0.961	0.953	0.955	0.999	0.995	0.978	0.979	0.987	0.981
			5	0.938	0.951	0.903	0.950	0.961	0.938	0.970	0.971	0.943	0.994	0.994	0.985	0.994	0.992	0.995
	30	(3, 28)	1	0.827	0.949	0.637	0.870	0.846	0.797	0.878	0.876	0.883	0.993	0.979	0.960	0.920	0.976	0.932
			5	0.563	0.965	0.460	0.840	0.919	0.941	0.871	0.945	0.958	0.980	0.969	0.958	0.943	0.977	0.961
		(10, 20)	1	0.943	0.950	0.936	0.945	0.947	0.948	0.963	0.952	0.951	1.000	0.997	0.990	0.973	0.992	0.975
			5	0.925	0.916	0.897	0.953	0.956	0.960	0.973	0.968	0.966	0.994	0.987	0.973	0.985	0.985	0.989
	50	(5, 46)	1	0.944	0.944	0.897	0.953	0.943	0.955	0.968	0.948	0.956	1.000	0.996	0.985	0.976	0.992	0.973
			5	0.843	0.946	0.677	0.927	0.915	0.923	0.944	0.924	0.947	0.991	0.978	0.976	0.953	0.985	0.961
		(17, 34)	1	0.936	0.934	0.942	0.946	0.947	0.954	0.953	0.951	0.957	1.000	1.000	0.999	0.979	0.999	0.981
			5	0.940	0.928	0.918	0.949	0.940	0.947	0.961	0.943	0.951	0.998	0.991	0.979	0.960	0.981	0.959
(1, 3, −2)	10	(3, 8)	1	0.956	0.989	0.975	0.897	0.866	0.866	0.919	0.894	0.898	0.999	1.000	0.981	0.981	0.990	0.987
			5	0.969	0.985	0.962	0.912	0.831	0.886	0.937	0.857	0.927	1.000	1.000	0.993	0.982	0.996	0.986
		(4, 7)	1	0.981	0.987	0.977	0.912	0.913	0.910	0.949	0.938	0.933	1.000	1.000	0.989	0.988	0.992	0.991
			5	0.979	0.986	0.970	0.882	0.925	0.912	0.924	0.947	0.932	1.000	1.000	0.983	0.985	0.987	0.992
	20	(3, 18)	1	0.875	0.967	0.945	0.896	0.898	0.928	0.896	0.919	0.941	0.998	1.000	0.965	0.989	0.982	0.983
			5	0.639	0.971	0.858	0.890	0.821	0.878	0.907	0.852	0.902	0.985	0.996	0.937	0.963	0.966	0.966
		(7, 14)	1	0.951	0.982	0.976	0.951	0.942	0.937	0.970	0.964	0.955	0.999	1.000	0.982	0.987	0.990	0.992
			5	0.933	0.950	0.946	0.962	0.949	0.940	0.979	0.968	0.956	0.995	0.999	0.982	0.984	0.992	0.986
	30	(3, 28)	1	0.874	0.956	0.945	0.901	0.923	0.940	0.886	0.935	0.948	0.995	1.000	0.968	0.994	0.979	0.987
			5	0.608	0.956	0.862	0.806	0.868	0.892	0.789	0.887	0.906	0.978	0.996	0.929	0.971	0.951	0.971
		(10, 20)	1	0.944	0.987	0.989	0.949	0.954	0.956	0.965	0.967	0.966	1.000	1.000	0.993	0.997	0.995	0.997
			5	0.931	0.948	0.943	0.952	0.952	0.949	0.974	0.971	0.958	0.996	1.000	0.968	0.996	0.982	0.996
	50	(5, 46)	1	0.949	0.959	0.988	0.957	0.952	0.957	0.970	0.958	0.964	1.000	1.000	0.986	0.992	0.992	0.990
			5	0.886	0.950	0.940	0.960	0.947	0.954	0.963	0.954	0.955	0.995	1.000	0.979	0.990	0.986	0.989
		(17, 34)	1	0.937	0.983	0.988	0.950	0.956	0.955	0.955	0.961	0.962	1.000	1.000	0.999	1.000	0.999	1.000
			5	0.947	0.943	0.938	0.956	0.950	0.947	0.966	0.963	0.954	0.999	1.000	0.984	0.998	0.988	0.998

Note: CM is the parametric method that is the current default of Joinpoint software, and EmpQ1 and EmpQ2 denote the empirical quantile methods based on u' and (4), respectively. The mean function is $\mu(x) = \beta_{1,0} + \beta_{1,1}x + (\beta_{2,1} - \beta_{1,1})(x - \tau_1)^+ + (\beta_{3,1} - \beta_{2,1})(x - \tau_2)^+$, where $a^+ = \max(a, 0)$, $\beta_{1,0} = 1$, $\beta_{1,1} = \log(1 + \mathrm{APC}_1/100)$, $\beta_{2,1} = \log(1 + \mathrm{APC}_2/100)$, $\beta_{3,1} = \log(1 + \mathrm{APC}_3/100)$, and the errors are independent and normally distributed with mean zero and variance σ^2.

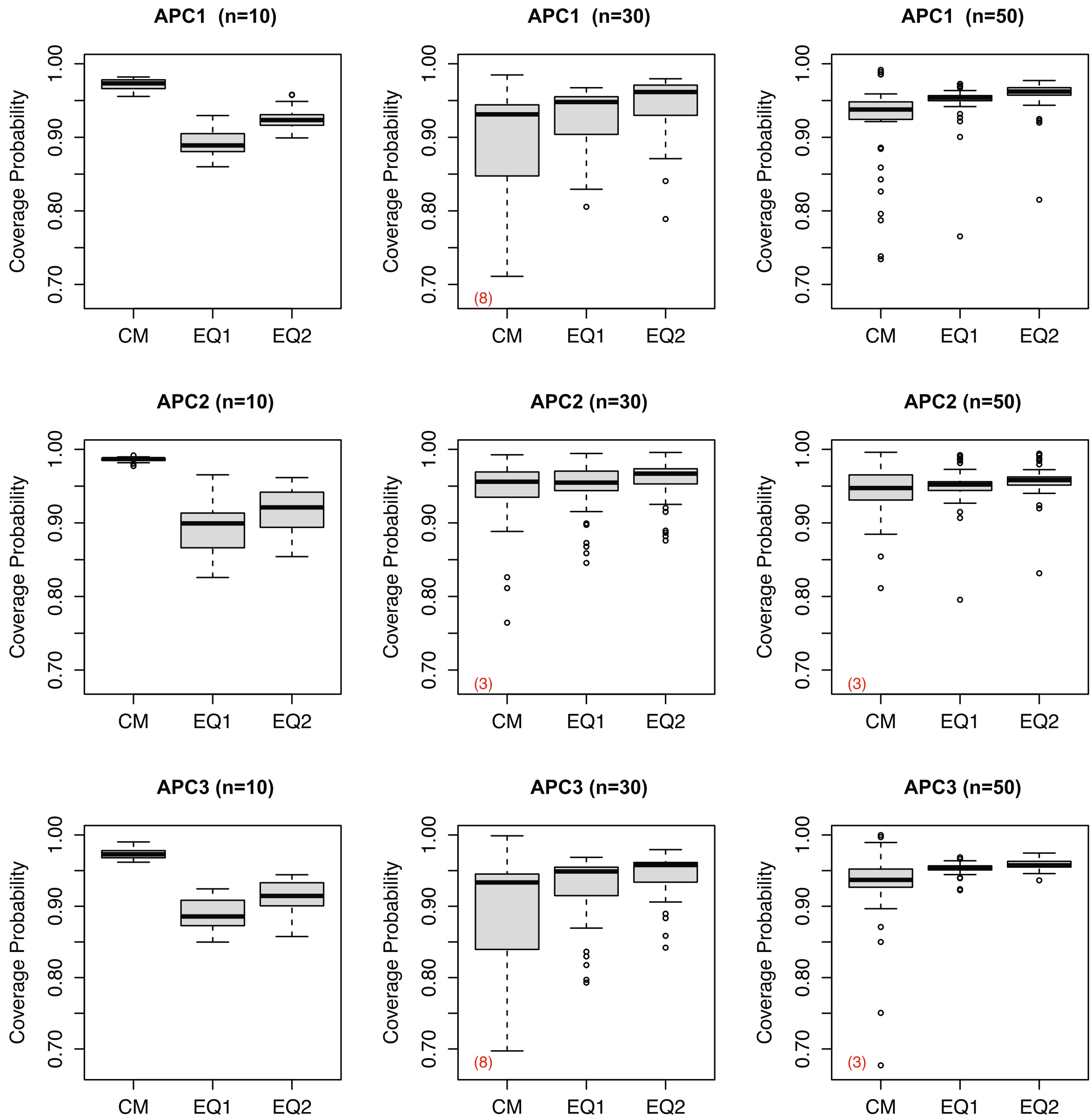


FIGURE 2 Box plots of coverage probability of APC confidence intervals by n: $\kappa = 2$, 38/39/43 cases for $n = 10/30/50$ (CM is the parametric method that is the current default of Joinpoint software, and EQ1 and EQ2 denote the empirical quantile methods based on u' and (4), respectively)

Although there were more cases included in our full results, we show only a few cases in Tables 1 and 2 that can illustrate major differences between the methods. Table 1 includes coverage probabilities of the 95% confidence intervals for the APC parameters constructed by three methods: the current parametric method based on the t-interval in (3) (CM), EmpQ1, and EmpQ2. For the τ parameters, the three methods presented are the likelihood based method in (5) (CM) and two empirical quantile methods. As indicated in Figures 1 and 2, the current parametric method (CM) produced coverage probabilities in a much bigger range compared to the empirical quantile methods, and sometimes coverage probabilities are as low as 0.4 and 0.5 (eg, 0.5074 for APC_1 when **APC**=(3,2), $n = 20$, $\tau = 3$, and $10\,000\sigma^2 = 5$, 0.4368 for APC_1 when **APC**=(3,2), $n = 30$, $\tau = 3$, and $10\,000\sigma^2 = 5$, 0.5760 for APC_2 when **APC**=(3,2), $n = 30$, $\tau = 27$, and $10\,000\sigma^2 = 5$). These are usually for the first segment APC with a very short first segment or for the second segment APC with a very short second segment. Such liberal tendency of the CM confidence intervals for APCs gets weaker when the change sizes are larger as indicated in the comparison of the APC value cases of (3, 2) and (3, 0). The performance gets worse as σ increases, and we don't observe a clear improvement with a larger n. Empirical quantile confidence intervals maintain the coverage probability at the nominal level of 95% much closer than the CM does. For the confidence intervals of τ, all of the three methods are usually conservative, sometimes too much, but it is a limitation with the annual grid search and the confidence interval based on the quarterly grid shows a slight improvement. Similar findings are observed in Table 2 where some results with $\kappa = 2$ are presented. Note that with 5000 simulations, the maximum margin of error in estimating the probability of correct selection is 0.007, and thus we report the results in three decimal places.

In summary, as stated in the Joinpoint site,[24] these simulation studies show that (i) the coverage probabilities of the EmpQ2 confidence intervals for the true APC and joinpoint are usually greater than those of the EmpQ1 confidence intervals, (ii) the coverage probabilities of the EmpQ1 confidence interval for the true APC were sometimes much below the nominal level, especially when the number of observations is small, and (iii) the EmpQ2 confidence interval for the true APC is sometimes conservative, but its coverage probabilities varied less than those of the EmpQ1 APC confidence interval. The empirical quantile confidence interval for the true APC is shown to be much more robust in terms of the segment length than the asymptotic parametric interval, the t-interval. Overall, the empirical quantile method outperforms the current parametric method to construct confidence intervals for the true APC values (ie, for the regression slope parameters), and we recommend the EmpQ2 method based on its less liberal behavior for data over a relatively short period of time and robustness. In the remaining sections, we use "EmpQ" to denote the EmpQ2 method.

4 | EXAMPLES

In this section, we apply the joinpoint regression model and Joinpoint software to study trend changes in some cancer sites. We first review the lung and bronchus cancer incidence and mortality rates for males and females separately, which illustrate a general use of the methodology and where we do not observe significant differences in the results between the current default setting of Joinpoint and the new methods discussed in this paper, WBIC for the model selection and the empirical quantile method for the inference on APC and AAPC values. As a second example, we further examine male mortality rates where we observed some changes in the number of segments estimated depending on the maximum number of joinpoints (JP_{max}) used.

According to the American Cancer Society, lung cancer is the second most common cancer for men and women in the United States, and it is the leading cause of the cancer death.[37] Figure 3 shows the SEER 9 delay adjusted incidence rates obtained from the SEER program of the US National Cancer Institute and the US mortality rates from the National Center for Health Statistics for males and females separately for the 43 years between 1975 and 2017. The SEER 9 program includes rates starting from 1975 and covers approximately 9.4% of the US population.[38] The delay adjusted rates provide more accurate trend analysis, and the adjustment was made using the delay-adjustment factors derived from the December 2018 North American Association of Center Cancer Registries submission.[39] Further information on the delay-adjusted rates can be found in Clegg et al[40] and Midthune et al.[41]

In Figure 3, we observe that lung cancer cases for males increased in the earlier period and then rapidly decreased from the early 1980s. The decrease in mortality followed, and a significant mortality decrease has been observed from the early 1990s. Differently from the male incidence trend that started to decrease from the early 1980s, the female incidence rates has increased until the middle of the 2000s, although the rate of increase significantly slowed down from the early 1990s. A significant decrease in mortality has started from the middle of the 2000s for females. As shown in Figure 3, the difference in the lung cancer incidence rates between males and females has narrowed over time, and the difference between the male and female in the age-adjusted incidence rates in 2017 is 10.7 per 100 000, while the biggest difference has been observed in 1980 as 67.8 per 100 000. Lung cancer incidence and mortality rates are largely driven by smoking rates, and the considerable lag from changes in smoking rates and an associated increase/decrease in lung cancer incidence. While smoking rates used to be much larger for men than women, for more recent birth cohorts (starting approximately for those born in 1960 and after) the rates have been more comparable. For more details of differences in lung cancer rates between males and females, see Egleston et al[42] and Holford et al.[43]

The joinpoint regression fits are shown in Figures 4 and 5, where incidence and mortality rates for males and females are plotted along with the joinpoint regression fits. We used the log-linear model whose mean function follows a joinpoint regression model, made the weighted least squares fit to estimate the regression parameters, and conducted the annual grid search to estimate the locations of joinpoints. The number of joinpoints were determined by the WBIC method presented in this paper with the maximum number of joinpoints set at 7 and the confidence intervals for the last segment APC, the last 5 observation AAPC, and the last 10 observation AAPC are presented in Table 3. For all four cohorts presented in Figures 4 and 5, there is no difference in the number of joinpoints estimated by the permutation procedure and the WBIC method and also we didn't observe a change in the significance of the three parameters considered (the last

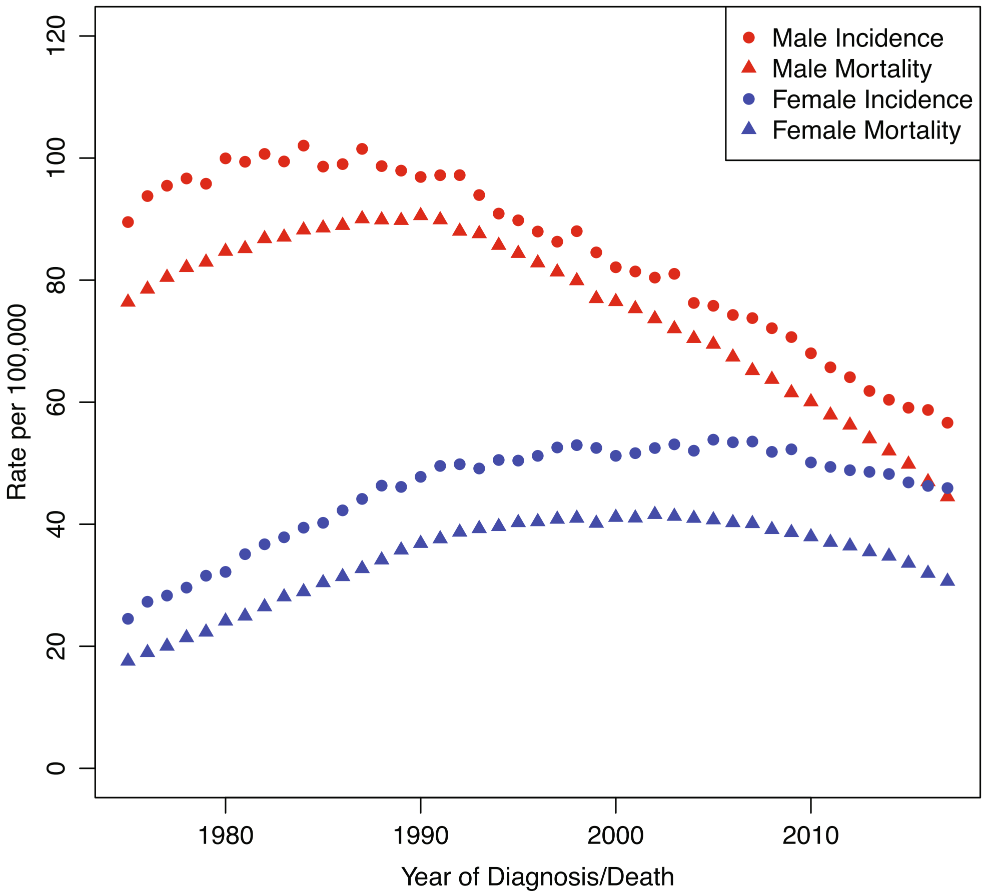
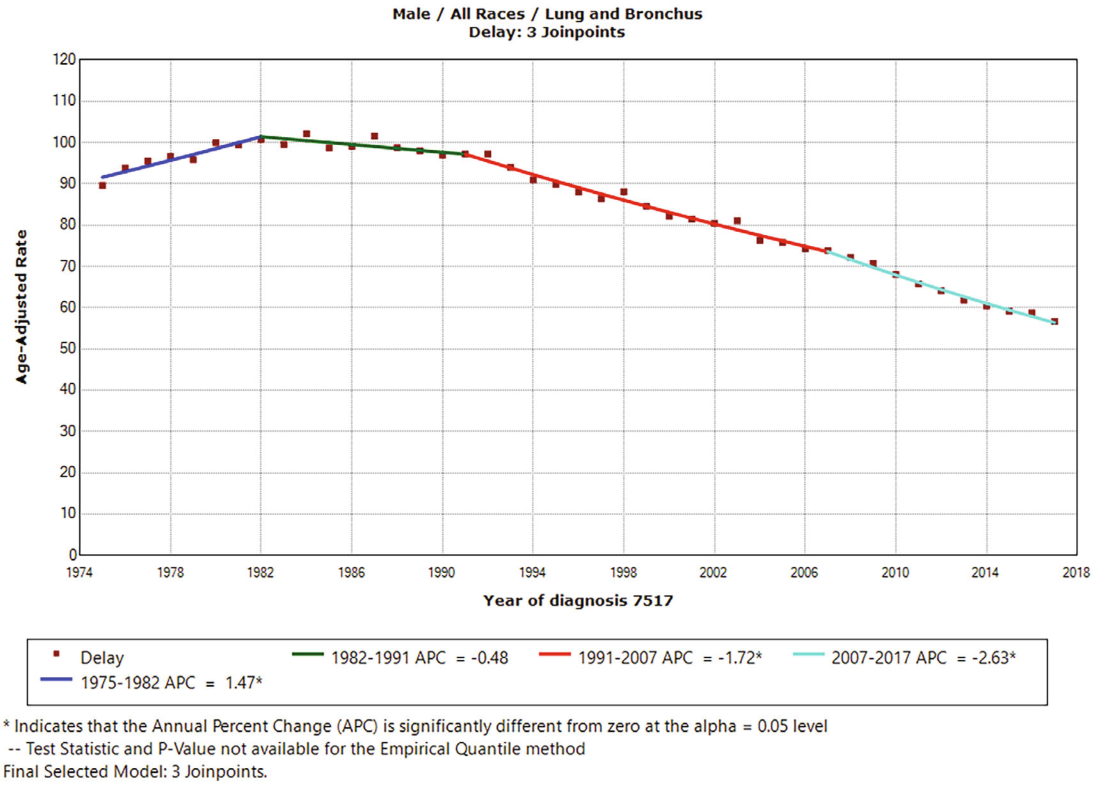


FIGURE 3 Lung and bronchus: SEER 9 delay-adjusted incidence and US death rates, 1975 to 2017

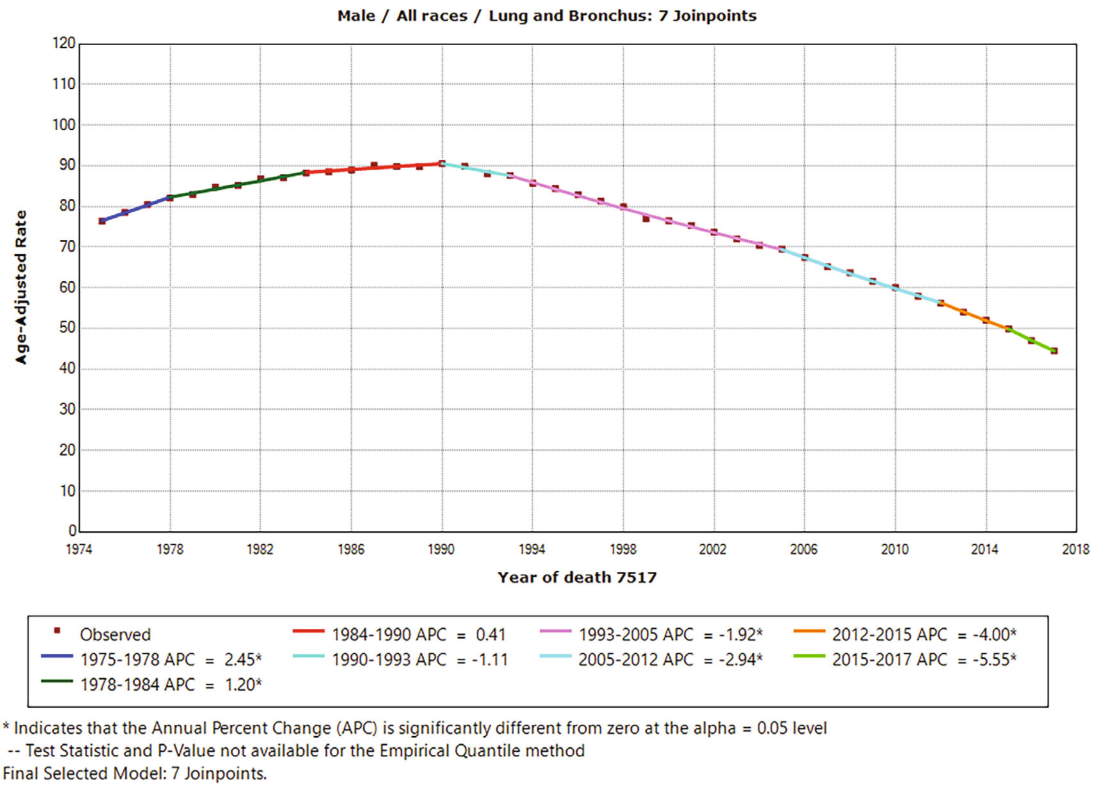
segment APC, the last 5 observation AAPC, and the last 10 observation AAPC) when the parametric confidence interval is compared to the EmpQ confidence interval. The 95% confidence intervals obtained by the parametric method (CM) and EmpQ are included in Table 3.

For male/female incidence trend, the length of the last segment was longer than 10 years, and thus we observed the same point estimates for the last segment APC and the last 5 and 10 year AAPCs. When the period over which the AAPC is calculated is composed with one segment, the parametric confidence intervals for the APC and AAPC are the same as shown in the male/female incidence cases, while the corresponding confidence intervals from the EmpQ are not always identical. It is because the length of the last segment in some resample data generated under the EmpQ method may not stay the same as the length of the last segment obtained for the original data and thus the last 5 (or 10) year period used to calculate the AAPC may have been composed with 2 segments in some resampled data. It would occur more frequently if the slope change between the two segments around the last joinpoint is not very big. For example, in the female incidence case, the last segment is from 2006 to 2017, and thus the point estimates for the last segment APC, the last 5 year AAPC, and the last 10 year AAPC are all equal. For female incidence rates, the EmpQ confidence intervals for the last segment APC and the last 5 year AAPC are identical, but the EmpQ confidence interval for the last 10 year AAPC shows a slight change from the other confidence intervals. For male incidence rates with last segment of (2007, 2017), the difference between the EmpQ confidence intervals for the last segment APC and last 5 year AAPC is minimal, and the EmpQ confidence interval for the last 10 year AAPC shows a more change from the other two confidence intervals, compared to the female incidence analysis. This can be explained by that the size of the APC change during the last two segments in male incidence rates is relatively smaller than the size of the APC change from the second segment to the third segment in female incidence rates. When the segment length is short as in the last segment of the male/female mortality rates, the parametric confidence intervals are usually wider than the EmpQ confidence intervals.

When we used the maximum number of joinpoints set as 5 or 6 ($JP_{max} = 5, 6$) for male mortality rate analysis, the final models selected by Perm and WBIC were different: 5 joinpoint model by Perm and 4 joinpoint model by WBIC. In Table 4 and Figure 6, we examine these fits made for the male mortality rates. For either fits made with $\hat{\kappa} = 4$ or 5, the

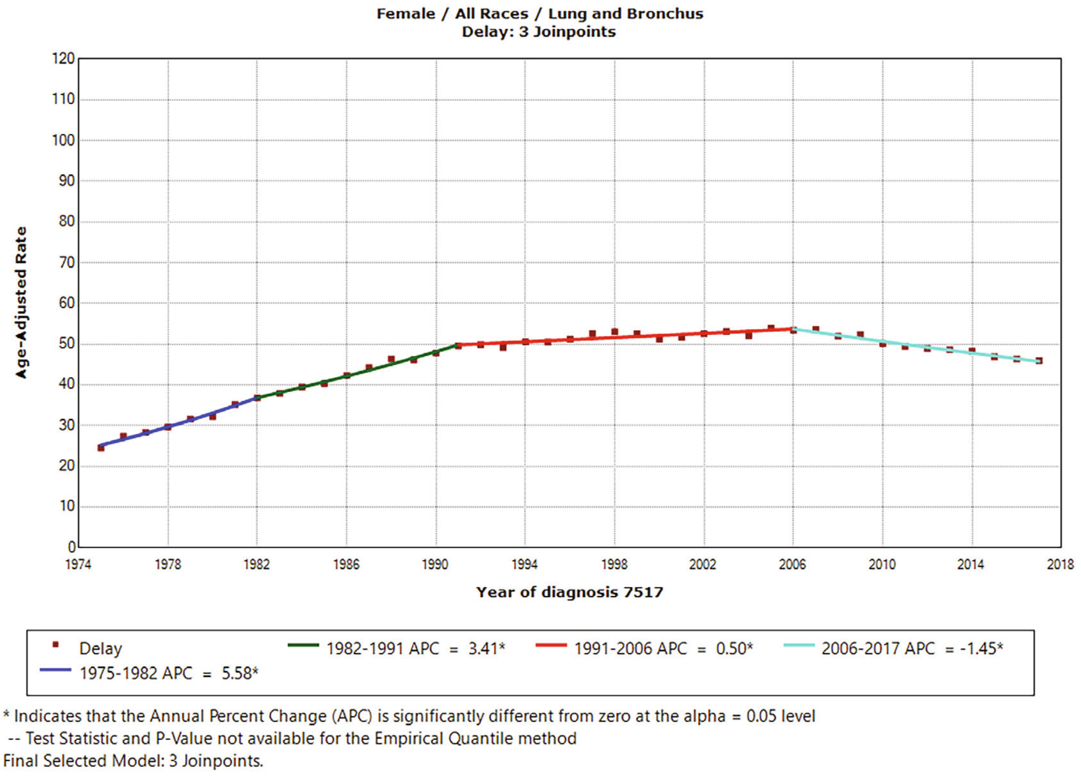


(A) Male Incidence

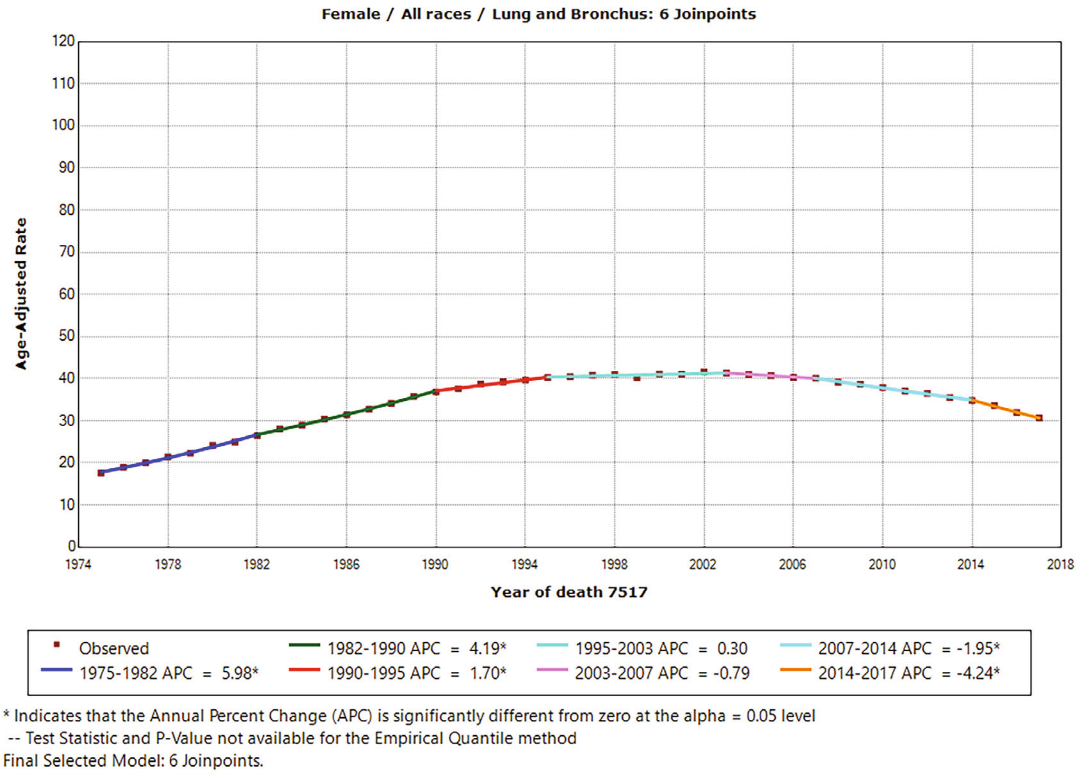


(B) Male Mortality

FIGURE 4 Lung and bronchus for males: SEER 9 delay-adjusted incidence and US death rates, 1975 to 2017



(A) Female Incidence



(B) Female Mortality

FIGURE 5 Lung and bronchus for females: SEER 9 delay-adjusted incidence and US death rates, 1975 to 2017

TABLE 3 Model fit and 95% confidence intervals for lung cancer incidence and mortality rates ($JP_{max} = 7$)

		Last	Last segment APC			Last 5 year AAPC			Last 10 year AAPC		
	NJP	Segment	PE	CM	EmpQ	PE	CM	EmpQ	PE	CM	EmpQ
Male incidence	3	(2007, 2017)	−2.630	(−2.924, −2.335)	(−3.155, −2.351)	−2.630	(−2.924, −2.335)	(−3.152, −2.352)	−2.630	(−2.924, −2.335)	(−2.965, −2.362)
Male mortality	7	(2015, 2017)	−5.553	(−6.947, −4.137)	(−6.449, −4.590)	−4.782	(−5.703, −3.852)	(−5.188, −4.528)	−3.879	(−4.412, −3.344)	(−4.055, −3.751)
Female incidence	3	(2006, 2017)	−1.447	(−1.686, −1.206)	(−1.715, −1.216)	−1.447	(−1.686, −1.206)	(−1.715, −1.216)	−1.447	(−1.686, −1.206)	(−1.714, −1.218)
Female mortality	6	(2014, 2017)	−4.242	(−5.117, −3.359)	(−5.250, −3.543)	−3.673	(−4.303, −3.039)	(−4.222, −3.352)	−2.718	(−3.056, −2.378)	(−2.948, −2.562)

Note: NJP is the number of joinpoints estimated by WBIC and PE denotes the point estimate. CM is the current default of Joinpoint software (parametric method for inference) and EmpQ denotes the empirical quantile method based on (4).

last segment was obtained as (2013, 2017), but the point estimates were slightly different, −4.837 for the 5 joinpoint model and −4.841 for the 4 joinpoint model, and the subsequent confidence intervals show small differences between the two methods: CM and EmpQ. The two graphs in Figure 6 are the joinpoint fits made for the male mortality rates with $\hat{\kappa} = 5$ and are included to illustrate different results we may get regarding the significance of the APC estimates. The significant APC values are marked with * for the fits with the parametric method (CM) in Figure 6A and the EmpQ method in Figure 6B. We note that the parametric method in Figure 6A indicates all APCs significant, but the third segment APC estimated as 0.345 for the period of (1984, 1991) is not significant under the EmpQ method. For the third segment APC, the 95% confidence intervals are obtained as (0.085, 0.606) and (−2.118, 0.578) by CM and EmpQ, respectively, and the lower limit of the CM confidence interval is only slightly larger than zero. Regarding the model selection, the permutation test of four joinpoints vs five joinpoints with $\text{JP}_{\text{max}} = 5$ estimated the P-value as 0.046, which is marginal, and the WBIC values were very close with 2.090 and 2.101 for $k = 4$ and 5, respectively.

5 | IMPACT OF PROPOSED CHANGES IN CANCER REPORTING

The surveillance, epidemiology, and the end results (SEER) program at the US National Cancer Institute had annually published the SEER Cancer Statistics Review (CSR) and they launched a new program, SEER*Explorer in 2021,[44] which is "an interactive website that provides easy access to a wide range of SEER cancer statistics." According to the SEER*Explorer site, the measures include incidence, mortality, prevalence, and survival statistics. The CSR 2020 report (Howlader et al[7]) includes statistics from 1975 to 2017 for the long term trend study, and the trend analysis for various cancer sites and by various cohorts are conducted using Joinpoint V 4.8. The Joinpoint default setting for the CSR 2020 analysis include the weighted least squares fitting using the standard error values provided by SEER*Stat software, the annual grid search to fit a joinpoint regression model, the permutation test for the model selection with 4499 permutations, 0.05 level of overall significance level (maximum over-fitting probability) and the maximum number of joinpoints set as 5, the t test and interval for the regression coefficients and APCs, and the parametric method for the joinpoint and AAPC confidence intervals.

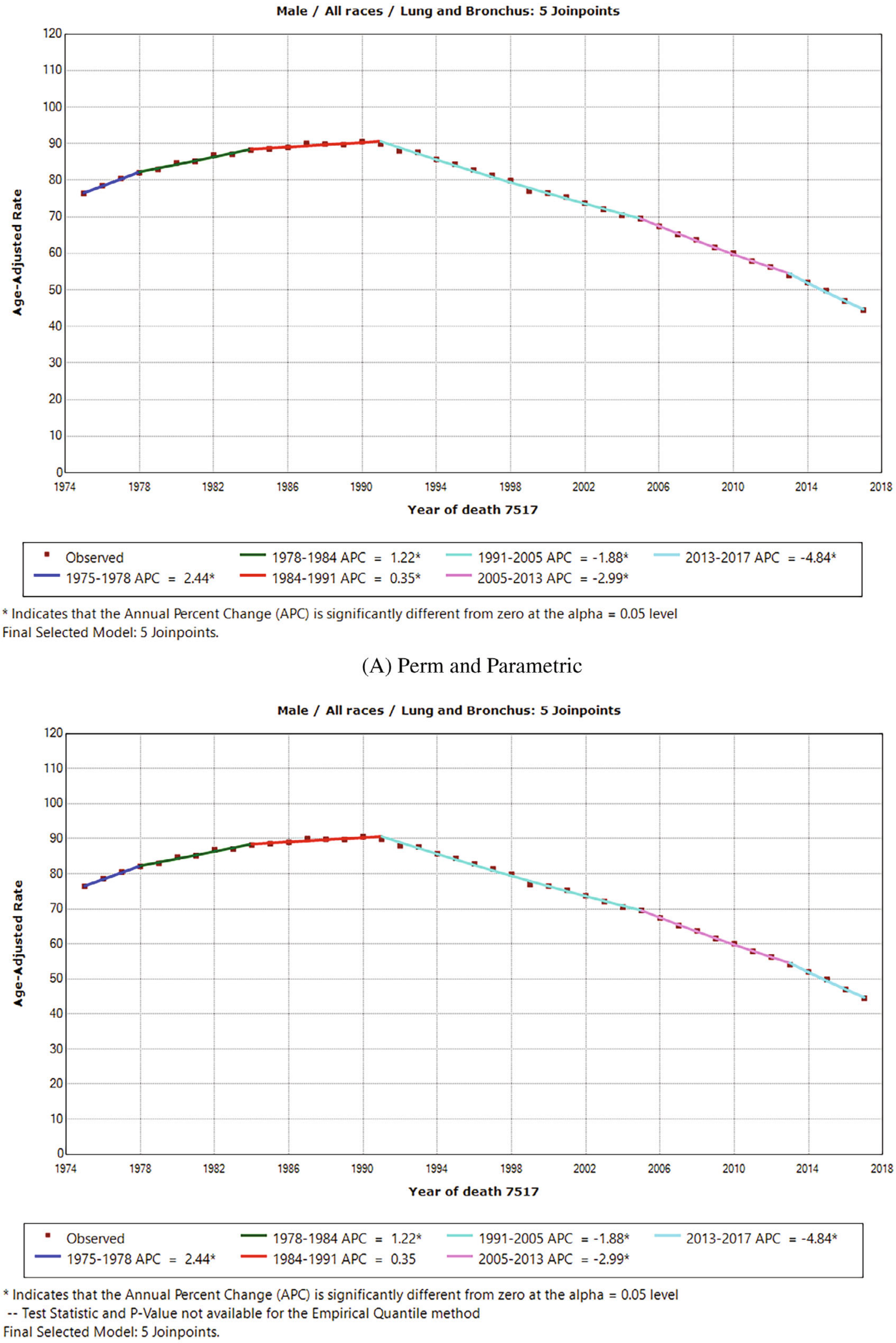
With a time period reaching to 44 years in SEER*Explorer to be available in 2022, there is a consideration of changing the default model selection method to the weighted BIC (WBIC) from the permutation procedure (Perm) and the default inference method for APC, AAPC, and joinpoints to the empirical quantile method (EmpQ) from the parametric method (Para). Main motivations of such changes are the computational burden with an increased maximum number of joinpoints in running the permutation test procedure and superior performance of the empirical quantile based inference for the model parameters. When we apply the new defaults of WBIC+EmpQ to the data available in 2022, the results will reflect both an additional data point as well as the effect of new statistical methods, and thus it is our interest to investigate and document how many results are different solely due to the changes in the methodology. Also, we are interested in verifying that results predicted from both theoretical considerations and simulation-based studies are borne out when applied to a large range of actual analysis. To address such questions, we compare the results for various cancer sites and cohorts obtained by using the current default setting (Perm+Para) with those obtained by using the empirical quantile methods for the APC, AAPC, and τ inferences and the weighted BIC method to select the model. To investigate how these method changes impact the results on the significance of the model parameters of our interests, for example, last segment APC and last 5 year AAPC, there can be various comparisons such as Perm+Para vs Perm+EmpQ, Perm+Para vs WBIC+Para, and so forth, and we will be presenting the results of Perm+Para (current default) vs Perm+EmpQ to assess the effects of a change in the inference method and Perm+Para (current default) vs WBIC+EmpQ to assess the combined effects of a change in both model selection and inference methods.

For the analysis summarized in Tables 5 to 10, we used the delay adjusted incidence rates provided by the SEER 9 program and the US mortality rates collected by the National Center for Health Statistics. Table 5 compares the results from the CM and EmpQ methods for the SEER 9 incidence rates, where CM is the current default method based on the permutation procedure (Perm) to select the number of joinpoints and the parametric method (Para) to construct the confidence intervals for the regression slope parameters and EmpQ is the empirical quantile method based on (4) called EmpQ2 in the previous section. We considered 198 cohorts for SEER 9 delay adjusted incidence rates that are formulated by various cancer sites, gender and race groups, and examined the significance of the last segment APC, the AAPC over the last 5 observations, and the AAPC over the last 10 observations. For each parameter, a 3 by 3 table is constructed based on the significance of the estimated parameter obtained by the Joinpoint V 4.9 default run (Para) and the run with the empirical quantile method (EmpQ). Note that for both inference methods, the model selection was performed using the

TABLE 4 Model fit and 95% confidence intervals for lung cancer US male mortality rates ($JP_{max} = 5$ or 6)

		Last	Last segment APC			Last 5 year AAPC			Last 10 year AAPC		
	NJP	Segment	PE	CM	EmpQ	PE	CM	EmpQ	PE	CM	EmpQ
Perm	5	(2013, 2017)	−4.837	(−5.308, −4.364)	(−5.430, −4.405)	−4.837	(−5.308, −4.364)	(−5.240, −4.393)	−3.816	(−4.402, −3.589)	(−3.996, −3.702)
WBIC	4	(2013, 2017)	−4.841	(−5.345, −4.335)	(−5.498, −4.381)	−4.841	(−5.345, −4.335)	(−5.269, −4.373)	−3.814	(−4.058, −3.570)	(−4.001, −3.703)

Note: NJP is the number of joinpoints estimated and PE denotes the point estimate. CM is the current default of Joinpoint software (parametric method for inference) and EmpQ denotes the empirical quantile method based on (4).



(A) Perm and Parametric

(B) Perm and EmpQ

FIGURE 6 Parametric vs EmpQ-Lung and bronchus male US mortality, 1975 to 2017

TABLE 5 Significance of APC and AAPC for SEER 9 incidence data (CM vs EmpQ)

		Last segment APC			Last 5 year AAPC			Last 10 year AAPC			
		EmpQ			EmpQ			EmpQ			
		sig+	NS	sig−	sig+	NS	sig−	sig+	NS	sig−	Total
All cancer sites											
	sig+	52	14	0	51	13	0	65	4	0	
CM	NS	0	55	2	0	57	4	3	41	1	198
	sig−	0	15	60	0	12	61	0	5	80	
Large cancer sites											
	sig+	5	2	0	3	2	0	3	2	0	
CM	NS	0	15	0	0	17	0	0	4	0	39
	sig−	0	5	12	0	4	13	0	0	30	
Medium cancer sites											
	sig+	32	10	0	33	9	0	45	2	0	
CM	NS	0	26	1	0	26	3	3	22	0	96
	sig−	0	6	21	0	4	21	0	4	20	
Rare cancer sites											
	sig+	15	2	0	15	2	0	17	0	0	
CM	NS	0	14	1	0	14	1	0	14	1	63
	sig−	0	4	27	0	4	27	0	1	30	

Note: The fits are made for 198 cohorts of SEER 9 delay-adjusted incidence rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), and EmpQ denotes the empirical quantile method based on (4) following the model selection by the permutation procedure. sig+, NS, and sig− denote the cases where the estimates are significantly negative, are not significantly different from zero and are significantly positive, respectively.

TABLE 6 Significance of APC and AAPC for US mortality data (CM vs EmpQ)

		Last segment APC			Last 5 year AAPC			Last 10 year AAPC			
		EmpQ			EmpQ			EmpQ			
		sig+	NS	sig−	sig+	NS	sig−	sig+	NS	sig−	Total
All cancer sites											
	sig+	109	5	0	109	3	0	110	4	0	
CM	NS	7	99	3	10	90	9	9	96	3	498
	sig−	0	19	256	0	14	263	0	19	257	
Large cancer sites											
	sig+	0	0	0	0	0	0	0	0	0	
CM	NS	0	6	0	0	0	0	0	5	1	55
	sig−	0	1	48	0	0	55	0	1	48	
Medium cancer sites											
	sig+	28	2	0	33	1	0	29	1	0	
CM	NS	1	16	2	1	13	2	3	14	1	127
	sig−	0	7	71	0	4	73	0	7	72	
Rare cancer sites											
	sig+	81	3	0	76	2	0	81	3	0	
CM	NS	6	77	1	9	77	7	6	77	1	316
	sig−	0	11	137	0	10	135	0	11	137	

Note: The fits are made for 498 cohorts of US mortality rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), and EmpQ denotes the empirical quantile method based on (4) following the model selection by the permutation procedure. sig+, NS, and sig− denote the cases where the estimates are significantly negative, are not significantly different from zero and are significantly positive, respectively.

TABLE 7 Significance of APC and AAPC for SEER 9 incidence data (CM vs WBIC+EmpQ)

Number of Joinpoints (CM→ WBIC)				Last segment APC WBIC+EmpQ sig+	NS	sig−	Last 5 year AAPC WBIC+EmpQ sig+	NS	sig−	Last 10 year AAPC WBIC+EmpQ sig+	NS	sig−	Total
All cancer sites													
Up	8		sig+	60	6	0	69	0	0	59	5	0	
Same	177	CM	NS	1	51	5	5	37	2	2	52	7	198
Down	13		sig−	0	7	68	0	0	86	0	2	71	
Large cancer sites													
Up	0		sig+	7	0	0	5	0	0	5	0	0	
Same	39	CM	NS	0	13	2	0	4	0	0	15	2	39
Down	0		sig−	0	5	12	0	0	30	0	2	15	
Medium cancer sites													
Up	5		sig+	36	6	0	47	0	0	37	5	0	
Same	81	CM	NS	1	25	1	5	20	0	2	24	3	96
Down	10		sig−	0	1	26	0	0	24	0	0	25	
Rare cancer sites													
Up	3		sig+	17	0	0	17	0	0	17	0	0	
Same	57	CM	NS	0	13	2	0	13	2	0	13	2	63
Down	3		sig−	0	1	30	0	0	31	0	0	31	

Note: The fits are made for 198 cohorts of SEER 9 delay-adjusted incidence rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), WBIC is the weighted BIC method for model selection, and EmpQ denotes the empirical quantile method based on (4). sig+, NS, and sig− denote the cases where the estimates are significantly negative, are not significantly different from zero and are significantly positive, respectively.

permutation test with $\alpha = 0.05$. When the parameter is noted as significant, we checked whether the estimated parameter is positive or negative. For the last segment APC, the AAPC over the last five observations, and the AAPC over the last 10 observations, there were no changes in the results for 84.4% of the cohorts (167 out of 198), 85.4% of cohorts (169 out of 198) and 93.4% of the cohorts (186 out of 198), respectively.

Table 5 also includes the similar 3 by 3 tables for the cancer sites categorized by their case numbers: large cancer sites, medium cancer sites, and rare cancer sites. We categorized the cancer site cohorts into these three categories based on the number of cases, where the sites with more than 40 cases per 100 000, between 7 and 40 cases per 100 000, and less than 7 cases per 100 000 are classified as large, medium and rare cancer (Large: all sites, breast, prostate, lung and bronchus, colon and rectum; Medium: melanoma, urinary bladder, non-Hodgkin lymphoma, kidney and renal pelvis, thyroid, corpus and uterus, leukemia, pancreas, oral cavity and pharynx, liver &IBD, stomach; Small: other sites). We have 39 large, 96 medium, and 63 rare sites for the incidence rate analysis. In all of the cases, more changes from CM significance to EmpQ nonsignificance than from EmpQ significance to CM significance were observed, and a higher proportion of changes from CM significance to EmpQ nonsignificance was more prominent for large cancer sites. Similar analysis are conducted for the US mortality data, and the results are summarized in Table 6. For the mortality analysis, the total of 498 cohorts were considered, among which 55, 127, and 316 cohorts belong to the large, medium, and rare cancer site groups, respectively. The proportions of the cohorts where the results of CM and EmpQ analysis are not different are slightly higher than those for the incidence cases, greater than 90% for any parameter of interest and for any cohort groups, but change proportions do not exhibit a pattern according to the large/medium/rate cancer site groups.

We also compared the combined effects of the weighted BIC (WBIC) and EmpQ on the significance of the last segment APC and the AAPCs over the last 5 and 10 observations, and summarized the results in Tables 7 and 8. The first two

TABLE 8 Significance of APC and AAPC for US mortality data(CM vs WBIC+EmpQ)

Number of Joinpoints (CM→ WBIC)				Last segment APC WBIC+EmpQ			Last 5 year AAPC WBIC+EmpQ			Last 10 year AAPC WBIC+EmpQ			
				sig+	NS	sig−	sig+	NS	sig−	sig+	NS	sig−	Total
All cancer sites													
Up	49		sig+	109	5	0	110	2	0	110	4	0	
Same	402	CM	NS	11	9	8	13	82	14	13	87	8	498
Down	47		sig−	0	23	252	0	17	260	0	23	253	
Large cancer sites													
Up	3		sig+	0	0	0	0	0	0	0	0	0	
Same	44	CM	NS	0	5	1	0	0	0	0	4	2	55
Down	8		sig−	0	1	48	0	0	55	0	1	48	
Medium cancer sites													
Up	18		sig+	27	3	0	33	1	0	28	2	0	
Same	93	CM	NS	5	11	3	4	8	4	7	9	2	127
Down	16		sig−	0	7	71	0	4	73	0	7	72	
Rare cancer sites													
Up	28		sig+	82	2	0	77	1	0	82	2	0	
Same	265	CM	NS	6	74	4	9	74	10	6	74	4	316
Down	23		sig−	0	15	133	0	13	132	0	15	133	

Note: The fits are made for 498 cohorts of US mortality rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), WBIC is the weighted BIC method for model selection, and EmpQ denotes the empirical quantile method based on (4). sig+, NS, and sig− denote the cases where the estimates are significantly negative, are not significantly different from zero and are significantly positive, respectively.

columns of Tables 7 and 8 list the number of cohorts where the number of joinpoints selected by WBIC is larger than (Up), same as (Same), or smaller than (Down) the one by the permutation test (CM) among all cohorts as well as in the three categories separately. We noted that 89.4% of cohorts in the SEER 9 analysis (177 out of 198 cohorts) didn't undergo a change in the final selected model when the permutation test and WBIC methods were used, while 8 (4.0%) and 13 cohorts (6.6%) showed a larger and fewer numbers of joinpoints selected by the WBIC method compared to the permutation test procedure, respectively. In the US mortality trend analysis, the number of joinpoints selected were same for the 80.7% of cohorts (402 out of 498) while the larger or smaller number of joinpoints was selected for the 9.8% (49 out of 498) and 9.4% (47 out of 498) of the cohorts, respectively.

Table 9 includes a summary according to the proportions of changes, from/to significance to/from nonsignificance between the CM setting (Perm+Para) and the WBIC+EmpQ method as well as between CM (Perm+Para) and EmpQ (Perm+EmpQ), observed for the SEER 9 incidence data. Although the WBIC+EmpQ method employs a different selection method as well as a different inference method, the lower proportions of changes in the significance of the APC, last 5 year AAPC and last 10 year AAPC from the CM results are observed in many cases compared to the changes of significance from the CM method (Perm+Para) to the Perm+EmpQ method. For the US mortality data analysis summarized in Table 10, we note that the WBIC+EmpQ method resulted in more changes in the significance of the parameters, either CM nonsignificance to WBIC+EmpQ significance or CM significance to EmpQ nonsignificance. Because the performances of each selection method (Perm or WBIC) and the inference method (parametric or empirical quantile method) depends on many factors such as variability in the data, lengths of segments, change size and so forth, we do not observe any specific pattern of changes in Tables 5 to 10. However, when we compare the current default method (Perm+Para) to the proposed method (WBIC+EmpQ), no change is observed in the significance of the last segment APC, last 5 year AAPC, and last 10 year AAPC for more than 90% of incidence and mortality cohorts analyzed.

TABLE 9 Proportion of changes in the significance of APC and AAPC for SEER 9 incidence data

		Last segment APC				Last 5 year AAPC				Last 10 year AAPC			
		Perm+EmpQ		WBIC+EmpQ		Perm+EmpQ		WBIC+EmpQ		Perm+EmpQ		WBIC+EmpQ	
		NS	Sig	NS	Sig	NS	Sig	NS	Sig	NS	Sig	NS	Sig
All cancer sites													
CM	NS	0.278	0.010	0.258	0.030	0.288	0.020	0.187	0.035	0.202	0.020	0.263	0.045
	Sig	0.146	0.566	0.066	0.646	0.126	0.566	0	0.778	0.045	0.732	0.035	0.657
Large cancer sites													
CM	NS	0.385	0	0.333	0.051	0.436	0	0.103	0	0.103	0	0.385	0.051
	Sig	0.179	0.436	0.128	0.487	0.154	0.410	0	0.897	0.051	0.846	0.051	0.513
Medium cancer sites													
CM	NS	0.265	0.010	0.260	0.021	0.271	0.031	0.208	0.052	0.229	0.031	0.250	0.052
	Sig	0.167	0.552	0.073	0.646	0.135	0.563	0	0.740	0.063	0.677	0.052	0.646
Rare cancer sites													
CM	NS	0.222	0.016	0.206	0.032	0.222	0.016	0.206	0.032	0.222	0.016	0.206	0.032
	Sig	0.095	0.667	0.016	0.746	0.095	0.667	0	0.762	0.016	0.746	0	0.762

Note: The fits are made for 198 cohorts of SEER 9 delay-adjusted incidence rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), WBIC is the weighted BIC method for model selection, and EmpQ denotes the empirical quantile method based on (4). Sig and NS denote the cases where the estimates are and are not significantly different from zero, respectively.

TABLE 10 Proportion of changes in the significance of APC and AAPC for US mortality data

		Last segment APC				Last 5 year AAPC				Last 10 year AAPC			
		Perm+EmpQ		WBIC+EmpQ		Perm+EmpQ		WBIC+EmpQ		Perm+EmpQ		WBIC+EmpQ	
		NS	Sig	NS	Sig	NS	Sig	NS	Sig	NS	Sig	NS	Sig
All cancer sites													
CM	NS	0.199	0.020	0.181	0.038	0.181	0.038	0.165	0.054	0.193	0.024	0.175	0.042
	Sig	0.048	0.733	0.056	0.725	0.034	0.747	0.038	0.743	0.046	0.737	0.054	0.729
Large cancer sites													
CM	NS	0.109	0	0.091	0.018	0	0	0	0	0.091	0.018	0.073	0.036
	Sig	0.018	0.873	0.018	0.873	0	1	0	1	0.018	0.873	0.018	0.873
Medium cancer sites													
CM	NS	0.126	0.024	0.087	0.063	0.102	0.024	0.063	0.063	0.110	0.031	0.071	0.071
	Sig	0.071	0.780	0.079	0.772	0.039	0.835	0.039	0.835	0.063	0.795	0.071	0.787
Rare cancer sites													
CM	NS	0.244	0.022	0.234	0.032	0.244	0.051	0.234	0.060	0.244	0.022	0.234	0.032
	Sig	0.044	0.690	0.054	0.680	0.038	0.668	0.044	0.661	0.044	0.690	0.054	0. 680

Note: The fits are made for 498 cohorts of US mortality rates for 1975 to 2017. CM is the current default of Joinpoint software (permutation procedure for model selection and parametric method for inference), WBIC is the weighted BIC method for model selection, and EmpQ denotes the empirical quantile method based on (4). Sig and NS denote the cases where the estimates are and are not significantly different from zero, respectively.

6 | ON-GOING AND FUTURE ENHANCEMENTS

Since its first release in 1998, there have been a wide range of enhancements and updates made in Joinpoint software, and Table 11 summarizes significant enhancements made in Joinpoint software since Joinpoint V 1.0.

A continuous effort is being made for further improvement, and below we discuss several ongoing projects. First, for a more accurate, efficient, and flexible fitting, we plan to investigate computationally more efficient fitting algorithms. The grid search and Hudson's algorithms implemented in Joinpoint software are rather computationally expensive, although their applications are straightforward and exact, and there are some approximation methods available in literature and we plan to study their applicability and accuracies in the trend analysis. Although a fine grid search or a continuous fitting such as the Hudson's algorithm provides a more accurate model fit, the basic grid search applied to annual data has a simple interpretation on the estimated location of the joinpoint and how to interpret the joinpoints estimated using a continuous fitting algorithm would require some conceptual discussion (eg, how to interpret $\hat{\tau} = 2010.3$).

One of recent extensions of Joinpoint software is to incorporate a general weight matrix to address heteroscedastic and correlated errors, which was motivated from the trend analysis with complex survey data. For example, when over-lapping primary sampling units are used for a certain time period, the observed responses are often correlated, more strongly than in cancer incidence and mortality rates which represent a census rather than a sample. One way to handle such data is to use a general weight matrix and conduct the weighted least squares analysis using the aggregated data where the summary statistics are calculated at each time point to summarize the entire survey population. As a next step of the trend analysis with complex survey data, we are investigating how to select the number of joinpoints and estimate the model parameters using the individual level survey data rather than the aggregated data. This task involves further work to incorporate the design aspect of the study, and our aim is to develop a modified model selection procedure and to provide point estimates along with their standard error estimates.

The third problem that we are planning to investigate is the effect of model selection on the subsequent inferences and interactive effects of the fitting, model selection and inference methods. The current inference procedures are based on the consistency of the model dimension, that is, the consistency of the number of joinpoints estimated, and no empirical studies have been conducted to study the effect of the under-estimated or over-estimated model dimension to the subsequent APC and AAPC inferences. Our goal is to conduct simulation studies to investigate effects of various model selection methods on the inferences followed, with which we hope to be able to recommend a model selection method that performs best in this regard. Also, it has been observed that a more accurate fitting such as a continuous fitting based on the Hudson's algorithm or a fine grid search could result in a different model selected, and thus it is our interest to conduct empirical study on interactive effects of these fitting, model selection and inference methods.

Another project that we are pursuing is the clustering of un-ordered groups. Kim et al[10] proposed a method to cluster ordered groups according to their model similarity and applied the proposed methods to the 19 age-group data that are

TABLE 11 Summary of significant enhancements in Joinpoint

	Model Fitting & Summary	Inference	Model selection	Extension
V 1.0 (1998)	Grid search	Parametric	Permutation procedure	
V 3.0 (2005)			BIC added	
V 3.1 (2007)	Hudson's continuous fitting		Early stopping method	
V 3.3 (2008)	AAPC added	Parametric CI for AAPC		
V 3.4 (2009)				Comparability test added
V 3.5 (2011)			Modified BIC added	
V 4.2 (2015)		Empirical quantile CI for AAPC		
V 4.4 (2017)				Jump model analysis added
V 4.6 (2018)		Empirical quantile CI	BIC_3 and DDS added	
		for APC and τ		
V 4.7 (2019)			WBIC added	

naturally ordered. Its natural extension is the problem to cluster un-ordered groups. Since the naive extension of the method used in the ordered group clustering is not practically feasible due to its intensive computation, we have explored a few other clustering methods, and it is our goal to propose an efficient and accurate method to cluster un-ordered groups as another extension of Joinpoint. Its possible application would be a geographical clustering where our goal is to cluster similar geographic regions, not necessarily the neighboring regions, into several clusters with similar trend changes.

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DATA AVAILABILITY STATEMENT

The SEER data that support the findings of this study are available on request at https://seer.cancer.gov/data/access.html.

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